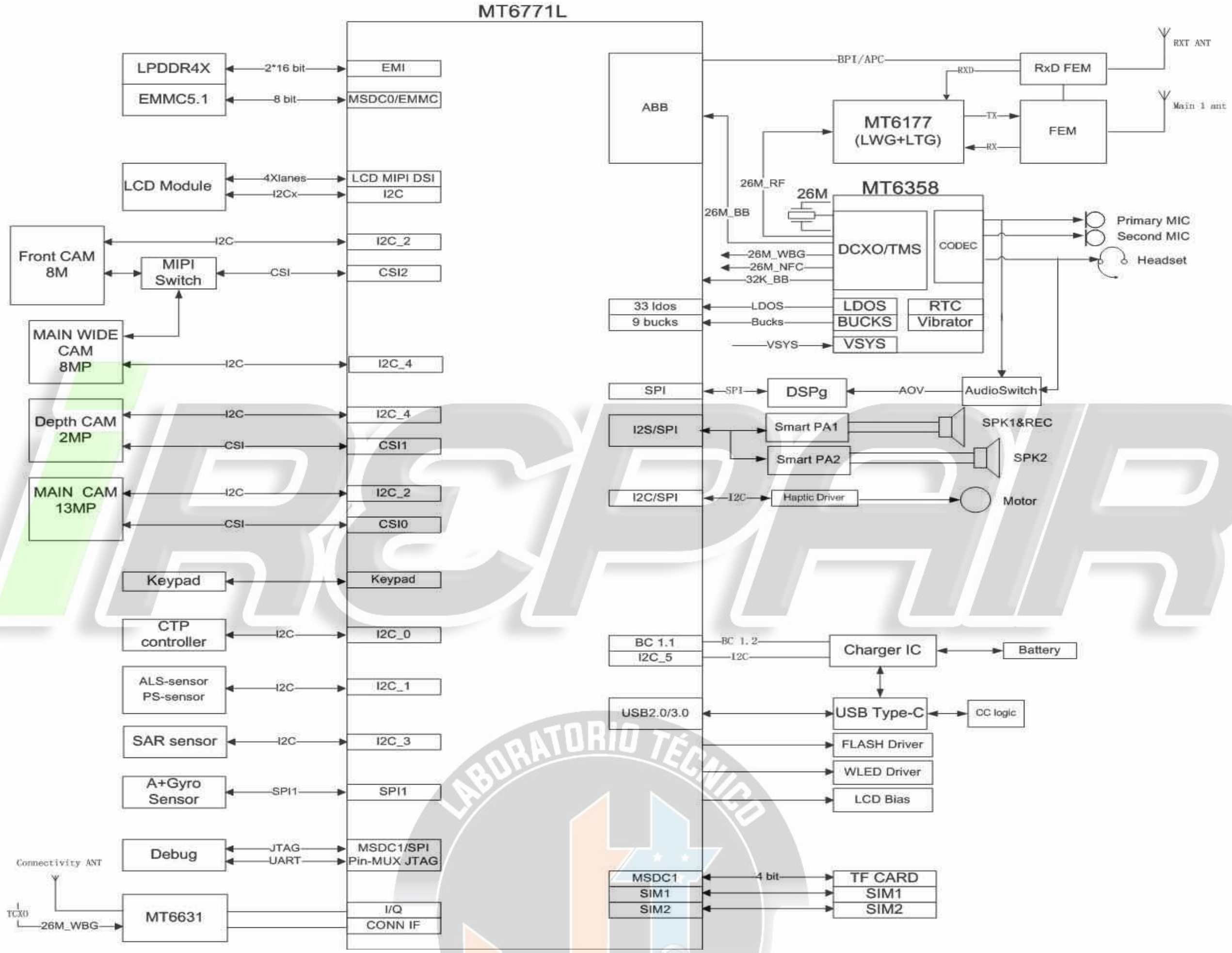
Riga L System Function Block Diagram



Title	Page Name = 00_Block Diagram	REV V10
DOCUMENT NO.:	Design Name = 98860_1_12_20190511120	Size D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ DESIGNER:	DESIGNER = YANGSHAO
WINGTECH		
Date:	Page Modify Date = Friday, May 10, 2019	Sheet 1 of 65

I2C	AP, SCP, SSPM	Function	I2C Spec.	Budget Timing	I2C Address
I2C-0	AP	CTP	400 Kbps	Yes.	ILI9881H I2C address: Write:0xAA, Read:0xAB
I2C-1(I3C)	AP ,SCP	ALS and P Sensor	400 Kbps	Yes.	TMD27504 I2C Address £°Write: 0x92, Read: 0x93 MN58406DKDN I2C Address £°Write: 0x82, Read: 0x83
		Ambient Light Sensor	400 Kbps	Yes.	TSL2560 I2C address: (Write:0x72, Read:0x73 MN26005UKDN I2C address: Write:0x92, Read:0x93
I2C-2 (I3C)	AP	Front Camera - 8M	400 Kbps	Yes.	S5K4H7YX03-FGX9 I2C address: Write:0x20, Read:0x21
		Main Camera - 13M	400 Kbps	Yes.	S5K3L6XX03-FGX9 I2C address: Write:0x5A, Read:0x5B
I2C-3	AP	HAPTIC DRIVER	400K bps	Yes.	DRV2624 I2C address: Write:0xB4, Read:0xB5 AW8624 I2C address: Write:0xB4, Read:0xB5
		Sar Sensor	400K bps	Yes.	A96T346DFP I2C address: Write:0x40, Read:0x41
		Smart PA1	400K bps	Yes.	TAS2563 I2C address: Write:0x9A, Read:0x9B
		Smart PA2	400K bps	Yes.	TAS2563 I2C address: Write:0x9C, Read:0x9D
I2C-4 (I3C)	AP	Wide Camera - 8M	400 Kbps	Yes.	S5K4H7YX03-FGX9 I2C address: Write:0x20, Read:0x21
		Depth Camera - 2M	400 Kbps	Yes.	OV02A10-GA4A I2C address: Write:0x7A, Read:0x7B
I2C-5	SSPM	Charger	400 Kbps	Yes.	RT9471D I2C address: Write:0xA6, Read:0xA7
		CC Logic	400 Kbps	Yes.	TUSB320 I2C address: Write:0x8E, Read:0x8F RT1711 address: Write:0x9C, Read:0x9D
		LCD Bias	400 Kbps	Yes.	SM5109 I2C address: Write:0x7C, Read:0x7D OCP2131 I2C address: Write:0x7C, Read:0x7D

Note : I2C Spec. : Standard mode (100 kbps) and Fast mode (400 kbps), Fast mode Plus (1 Mbps) and High-speed mode (3.4 Mbps)

SPI

SPI List

	GPIO	Device	Device IC	IC供应商
SPI0	GPIO[85;86;87;88]	Fingerprint	FPC1510A	
			/	/
SPI1	GPIO[161;162;163;164]	A+G	ICM-40607	INVENSENSE
			BMI160	BOSCH
SPI2	GPIO[0;1;2;94]	LCM	ILI9881H	ILITEC
SPI3	GPIO[21;22;23;24]	CODEC(DSP)	DBMD4	DSPG
SPI4	GPIO[4;5;6;7]			

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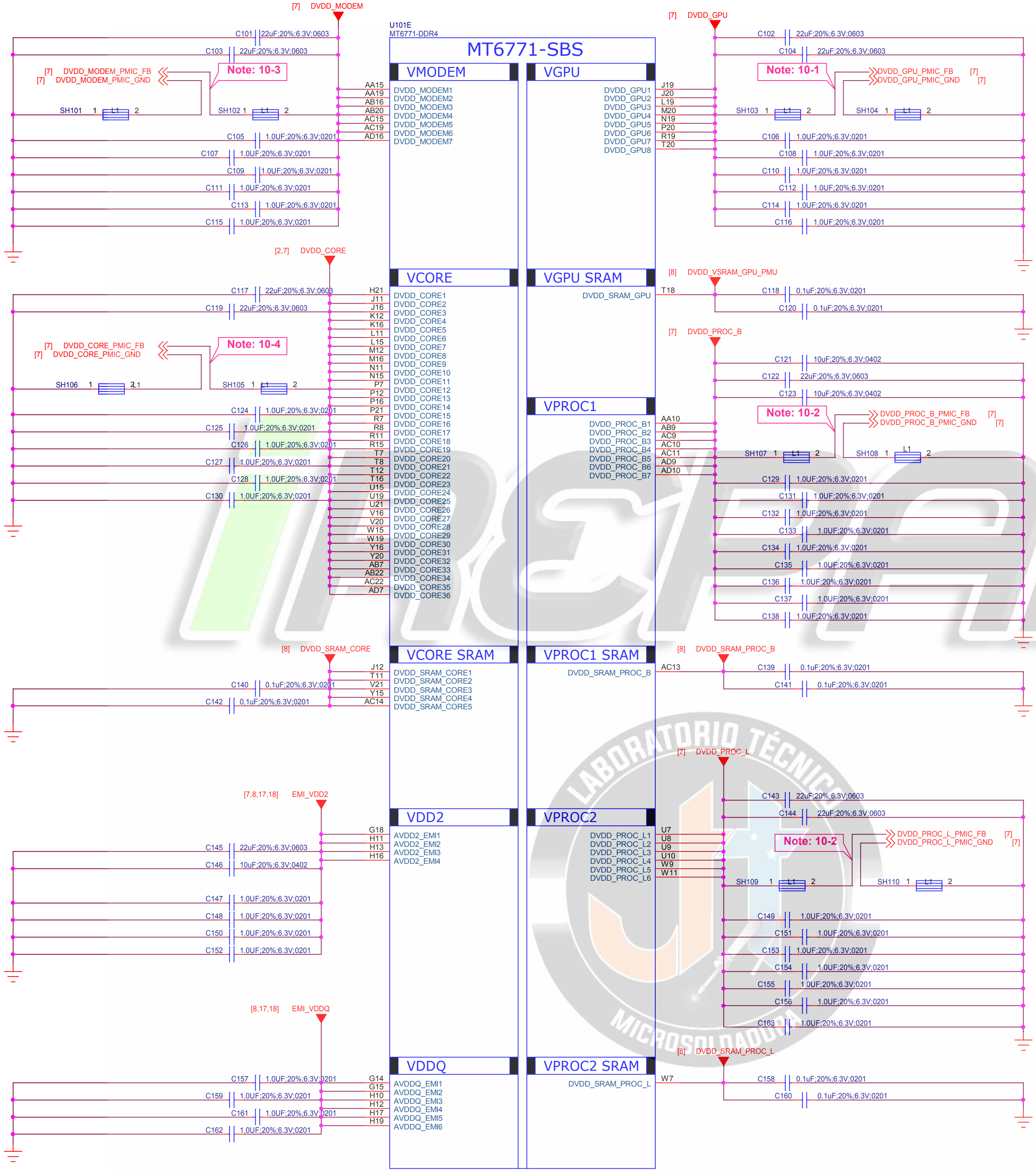
Title	Page Name = 00_I2C_SPI_Overview	REV : V10	
DOCUMENT NO.:	Design Name = 98860_1_12_201905111250	Size	D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = YANGSHAO
WINGTECH			
Date:	Page Modify Date = Friday, May 10, 2019	Sheet 2	of 65



Schematic design notice of "10_BB_POWER_PDN" page.

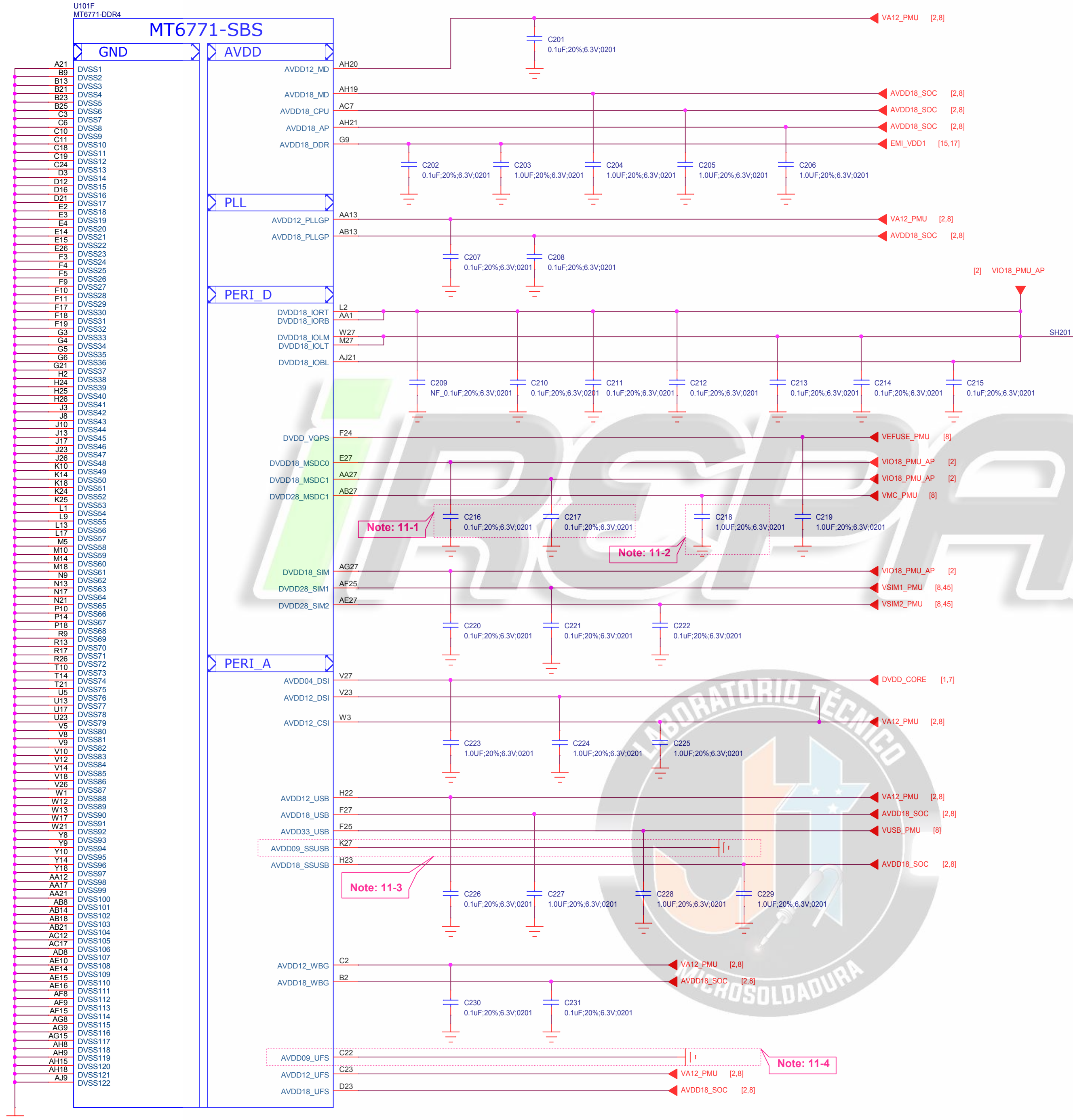
Note 10-1: Differential pair of DVDD_GPU remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

Note 10-2: Differential pair of DVDD_PROC remote sense must be close to BB's ball
Remote sense trace with GND shielding to PMIC (Differential)

Note 10-3: Differential pair of DVDD_MODEM remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

Note 10-4: Differential pair of DVDD_CORE remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

Title	01_BB_POWER_PDN	REV: V10	
DOCUMENT NO.:	Design Name	Size C	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Friday, May 10, 2019	Sheet 3 of 65	



Schematic design notice of "11_BB_POWER_IO" page.

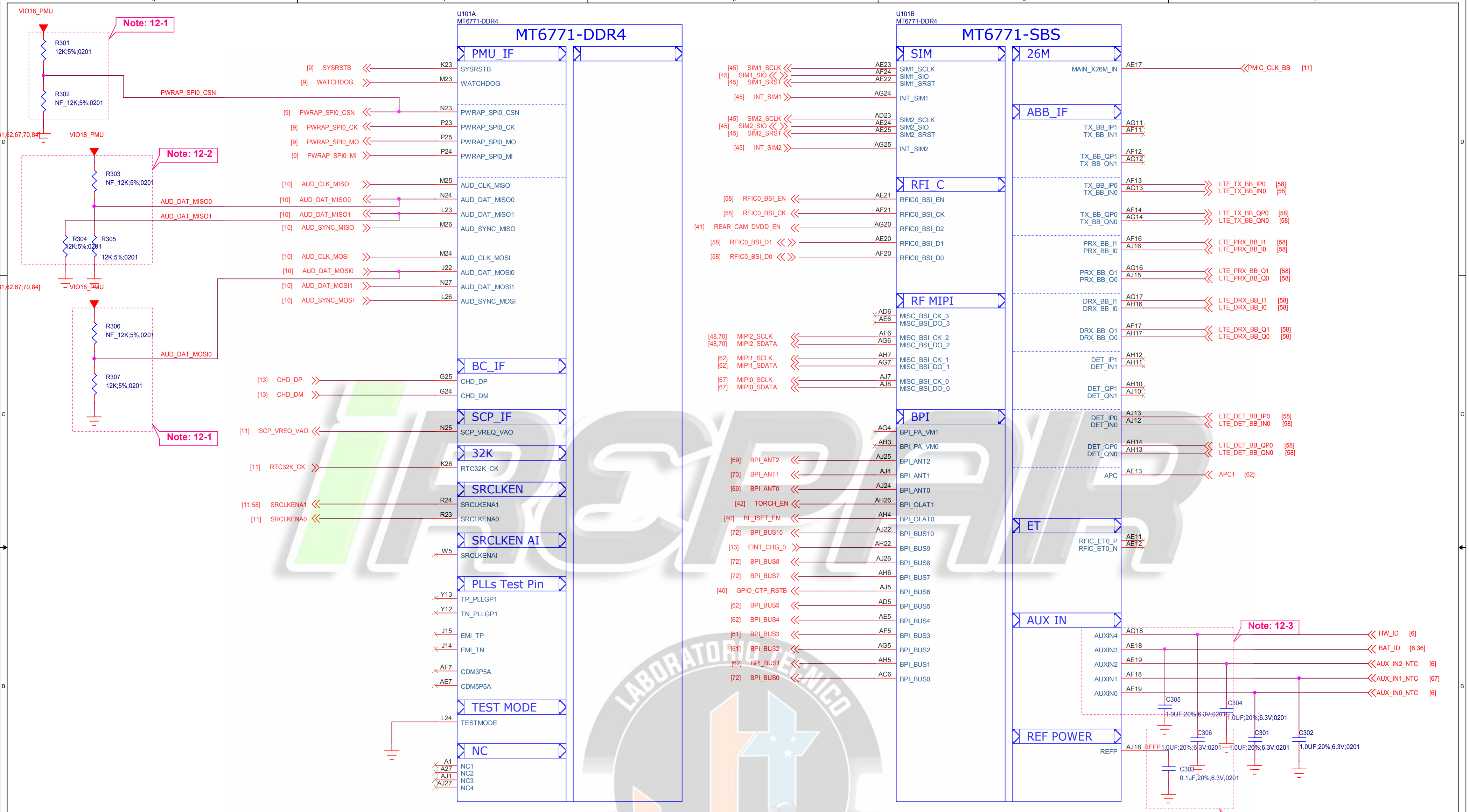
Note 11-1: C216 closed DVDD18_MSDC0 150mil
C217 closed DVDD18_MSDC1 150mil

Note 11-2: C218 closed DVDD28_MSDC1 150mil

Note 11-3: Connects "AVDD09_SSUSB" to GND when USB3.0 is not used.

Note 11-4: Connects "AVDD09_UFS" to GND when UFS is not used.

Title	02_BB_POWER_IO		REV: V10
DOCUMENT NO.:	Design Name		Size C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Friday, May 10, 2019	Sheet 4 of 65	



Schematic design notice of "12_BB_1" page.

Note 12-1: "PWRAP_SPI0_CSN" and "AUD_DAT_MOSI0" are bootstrap pins to select which interface will be the JTAG pin out.

PWRAP_SPI0_CSN	AUD_DAT_MOSI0	AP_JTAG	IO_JTAG
HI	LO	N/A	N/A
HI	HI	SPI_CSB/SPI_CLK/ SPI_MO/SPI_MI/EINT8	N/A
LO	LO	SPI_CSB/SPI_CLK/ SPI_MO/SPI_MI/EINT8	DPI_11/DPI_HSYNC/DPI_VSYNC/DPI_DE/ DPI_CK/DPI_D8/DPI_D9
LO	HI	MSDC1_CLK/CMD/ DAT0/DAT1/DAT2	N/A

Note 12-2: "AUD_DAT_MISO0" is bootstrap pin to enable serial JTAG output over USB2.0 interface or not.
When "AUD_DAT_MISO0" is pulled to high in system start up and then USB2.0 interface will be switched into serial JTAG mode.

"AUD_DAT_MISO1" is bootstrap pin to select system booting up from eMMC or UFS device.

AUD_DAT_MISO1	Booting device
LO	eMMC
HI	UFS

Note 12-3: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

Note 12-4: The de-coupling cap. for REFP (AJ18 ball) have to be placed as close to BB as possible.

Note 12-5: AUD_SYNC_MISO and AUD_CLK_MISO are DDR type feature in bootstrap

AUD_SYNC_MISO	AUD_CLK_MISO	DDR
LO	LO	LPDDR4X
LO	HI	LPDDR4X(Ext x 2 EN)
HI	LO	LPDDR3
HI	HI	LPDDR4X(Ext x 1 EN)

Title	03_BB_1_RF&SIM_IF	REV: V10	
DOCUMENT NO.:	Design Name	Size C	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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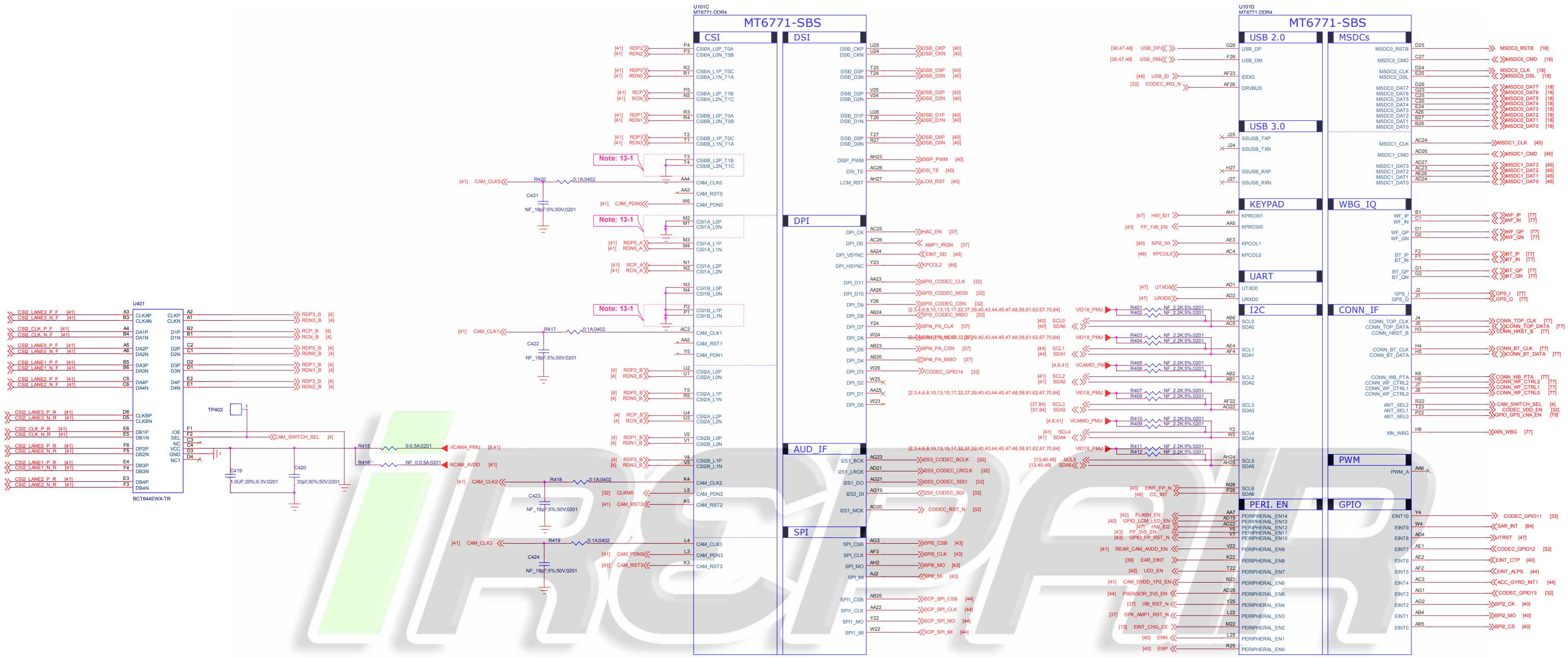

Schematic design notice of "13_BB_2" page.

Note 13-1: CSI ports which are no use could be connected to GND or set in NC.
For detail information, please refer to MT6771 Design Notice

Title	04_BB_2_MIPI&GPIO	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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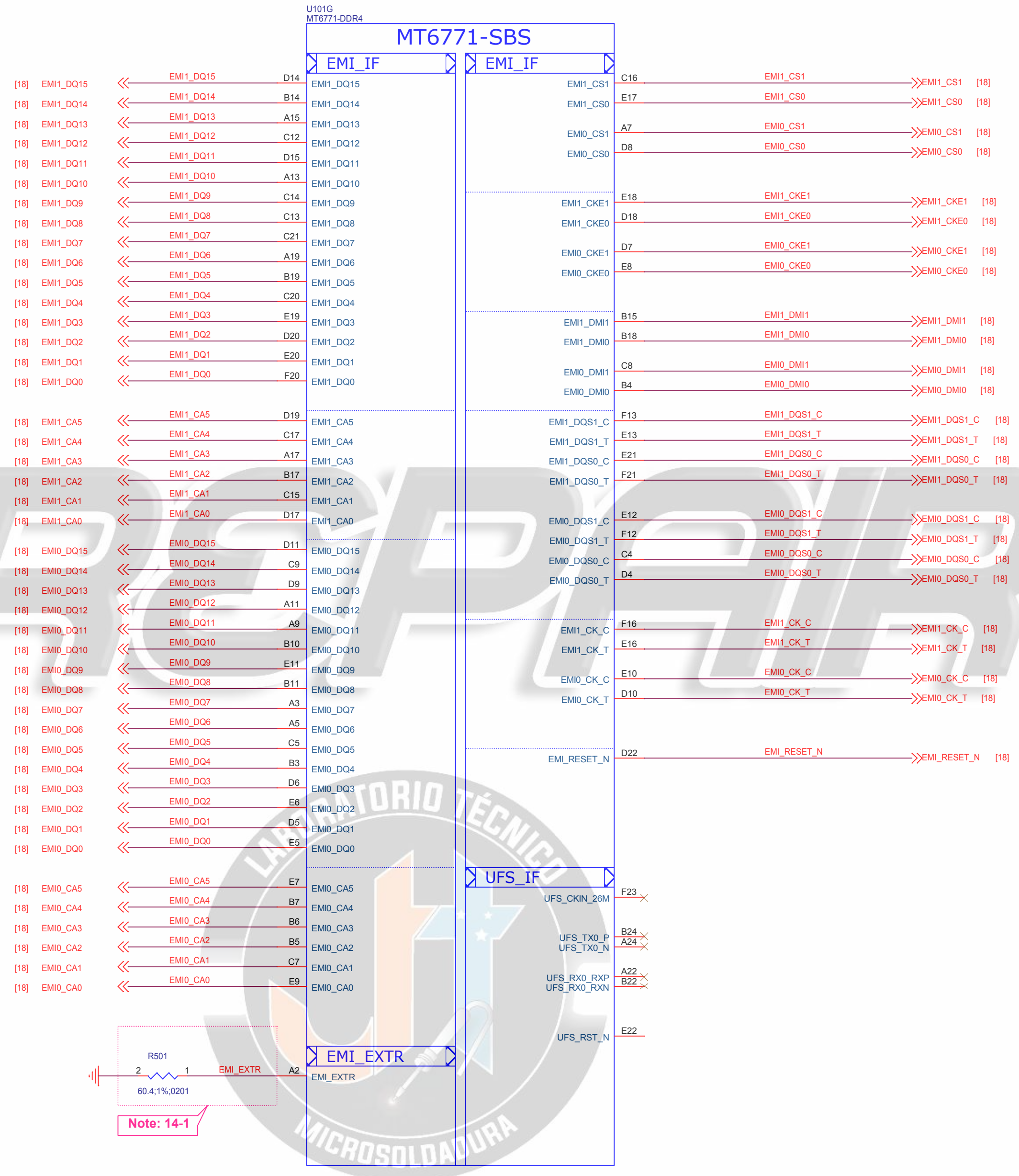

Schematic design notice of "14_BB_3" page.

Note 14-1: The resistor of EMI_EXTR for DRAM has to be placed near to BB as close as possible
R501, please select 60.4 ohm 1% resistor

Title	10_BB_ POWER_PDN		REV: V10
DOCUMENT NO.:	Design Name		Size C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Friday, May 10, 2019	Sheet 7 of 65	

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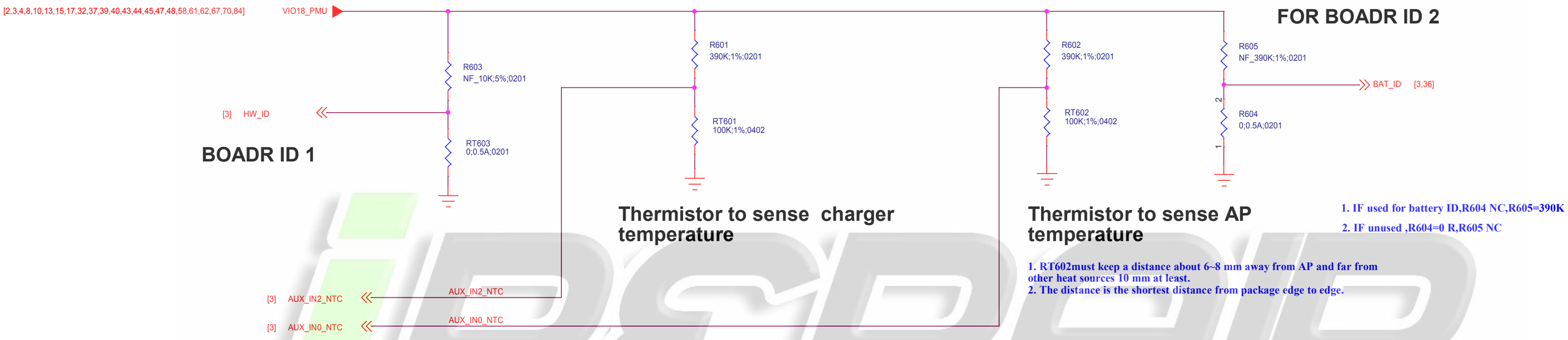

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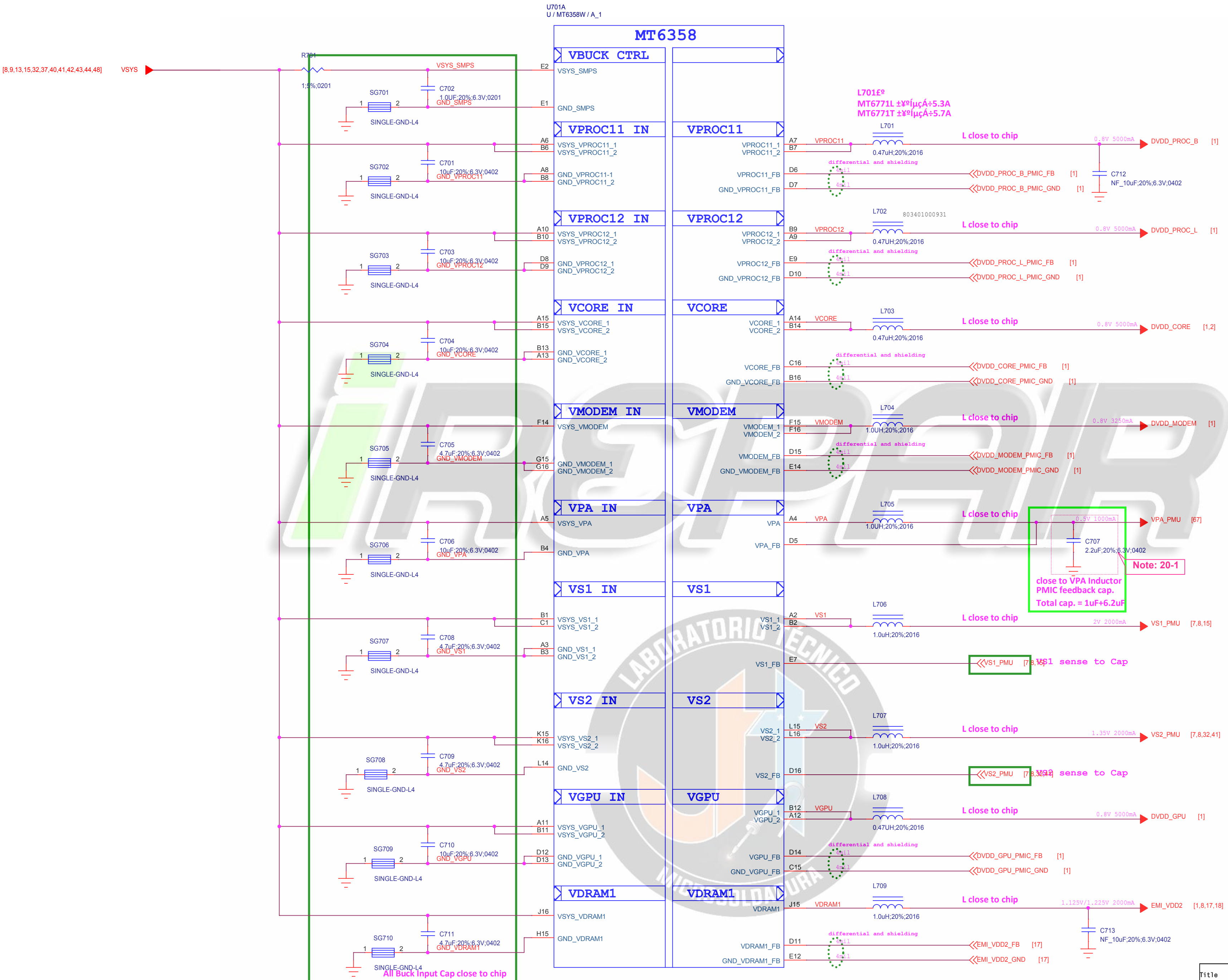
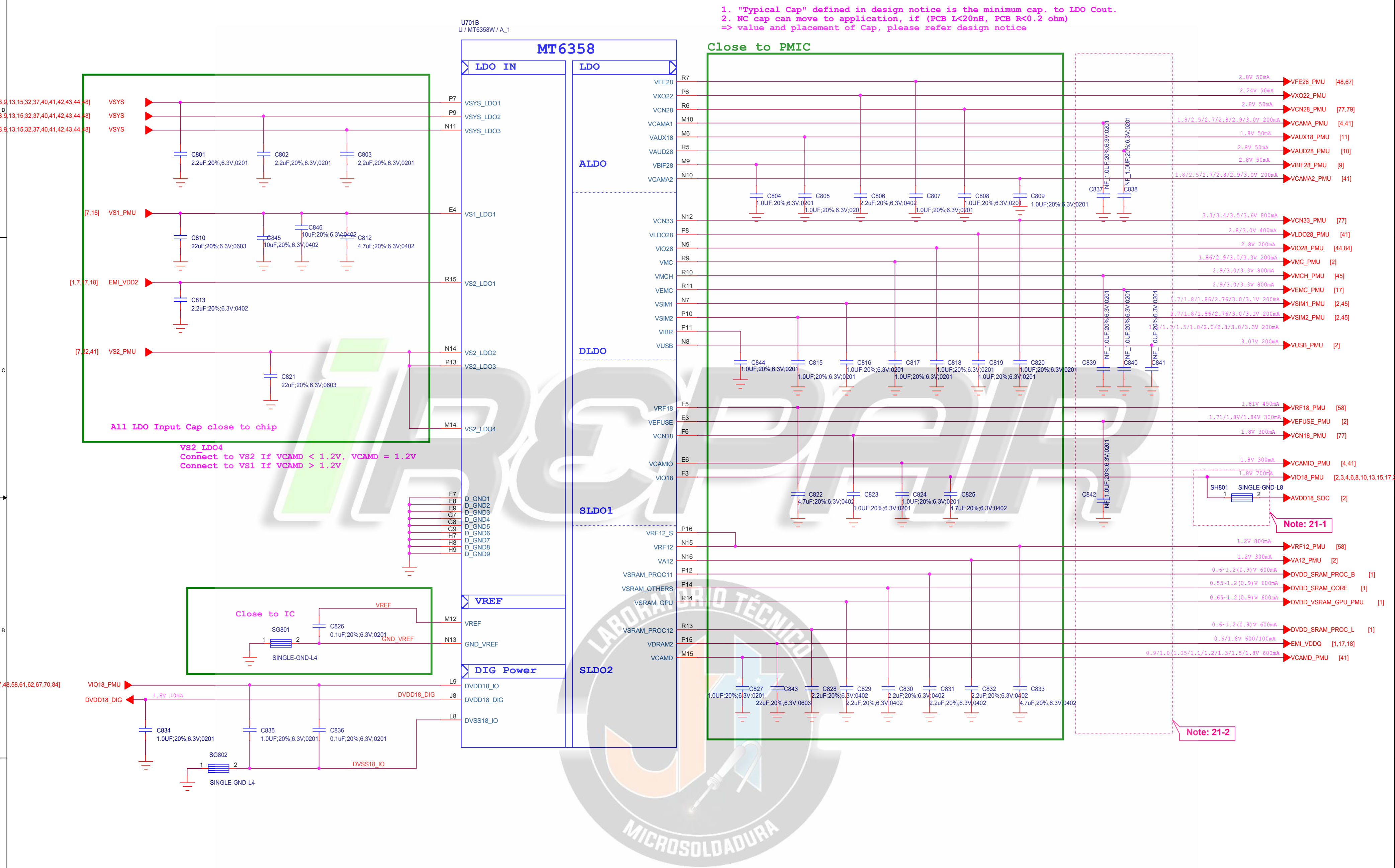
Title	06_BB_AUXADC_Thermal	REV: V10	
DOCUMENT NO.:	Design Name	Size C	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Friday, May 10, 2019	Sheet 8 of 65	



Schematic design notice of "20_POWER_MT6358-Buck" page.

Note 20-1: Please select C707 with 0402 size

Title	07_POWER_MT6358-Buck	REV: V10	
DOCUMENT NO.:	Design Name	Size C	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Friday, May 10, 2019	Sheet 9	of 65

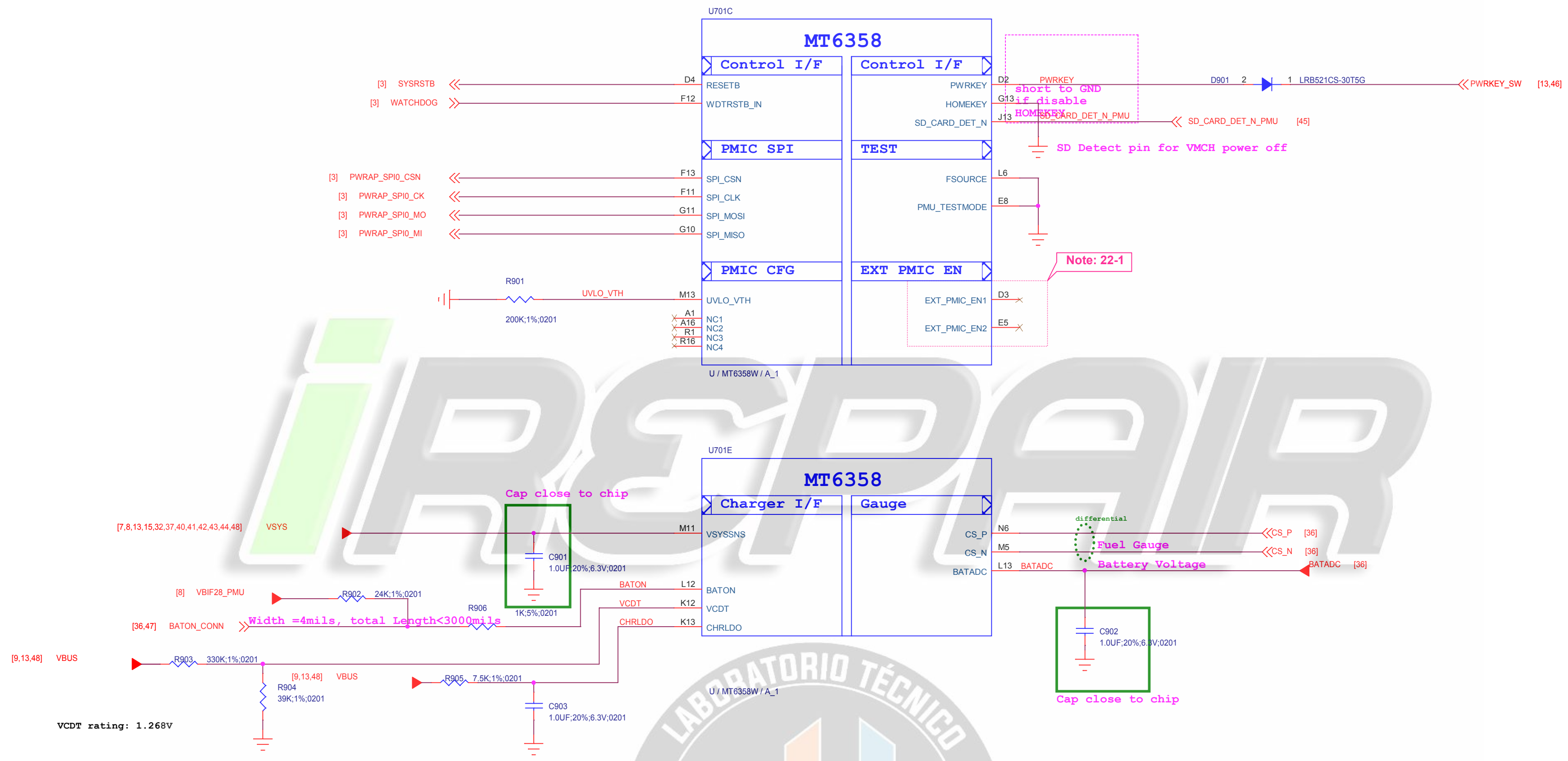


Schematic design notice of "21_POWER_MT6358-LDO" page.

Note 21-1: Please set SH801 close to C825, making star connection between VIO18_PMU and AVDD18_SOC near to LDO cap. C825
Please also refer to MT6358 design notice for further detail design information

Note 21-2: If these power trace can meet LDO layout constraint, these CAP can be NC or removed.
Please refer to MT6358 design notice.

Title	08_POWER_MT6358-LDO	REV: V10	
DOCUMENT NO.:	Design Name	Size C	
DEPARTMENT: WINGTECH-SH		DESIGNER:	LIUFENGLEI
WINGTECH			
Date: Friday, May 10, 2019		Sheet 10 of 65	



Schematic design notice of "22_POWER_MT6358-General"

Note 22-1: EXT_PMIC_EN1 : For UFS_1V8, and keep floating if it is not used
EXT_PMIC_EN2 : For VA09 of SSUSB/UFS, and keep floating if it is not used

Title	09_POWER_MT6358-General	REV:	V10
DOCUMENT NO.:	Design Name	Size	C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Friday, May 10, 2019	Sheet 11 of	65

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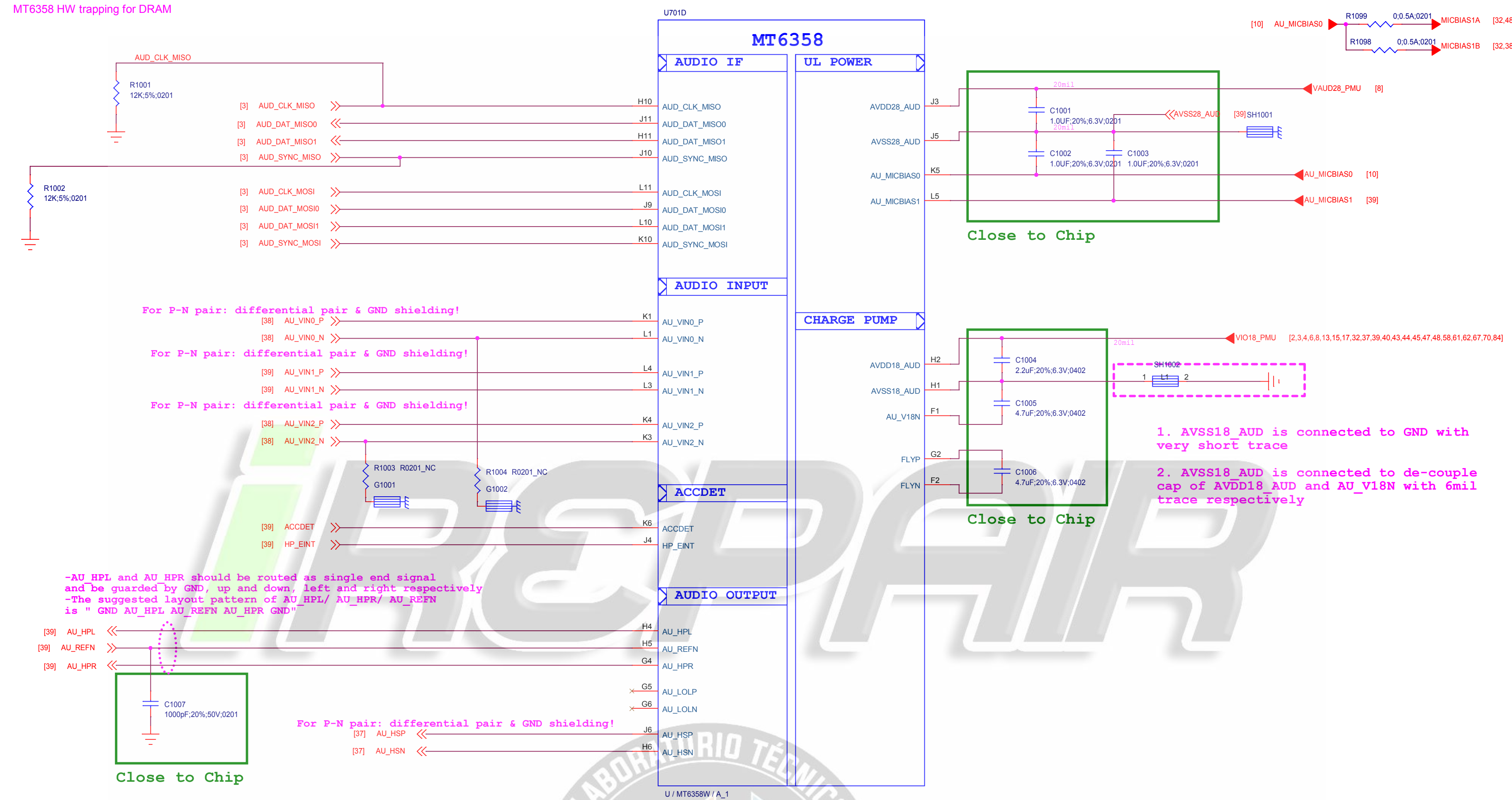

Schematic design notice of "23_POWER_MT6358-Audio"

Note 23-1: VDRAM 2 / VDRAM1 output voltage vs. trap pin.

HW GPIO configuration		Trapping Option		DRAM type	VDRAM2 Power source (VS2_LDO1_ball)
AUD_SYNC_MISO	AUD_CLK_MISO	VDRAM1	VDRAM2		
0	0	1.125V	0.6V	LP4X	VDRAM1
0	1	OFF	1.8V	LP4X (Ext x 2 EN)	VS1
1	0	1.225V	OFF	LP3	VDRAM1
1	1	1.125V	1.8V	LP4X (Ext x 1 EN)	VS1

Title	10_POWER_MT6358-Audio	REV: V10	
DOCUMENT NO.:	Design Name	Size	C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Friday, May 10, 2019	Sheet 12	of 65

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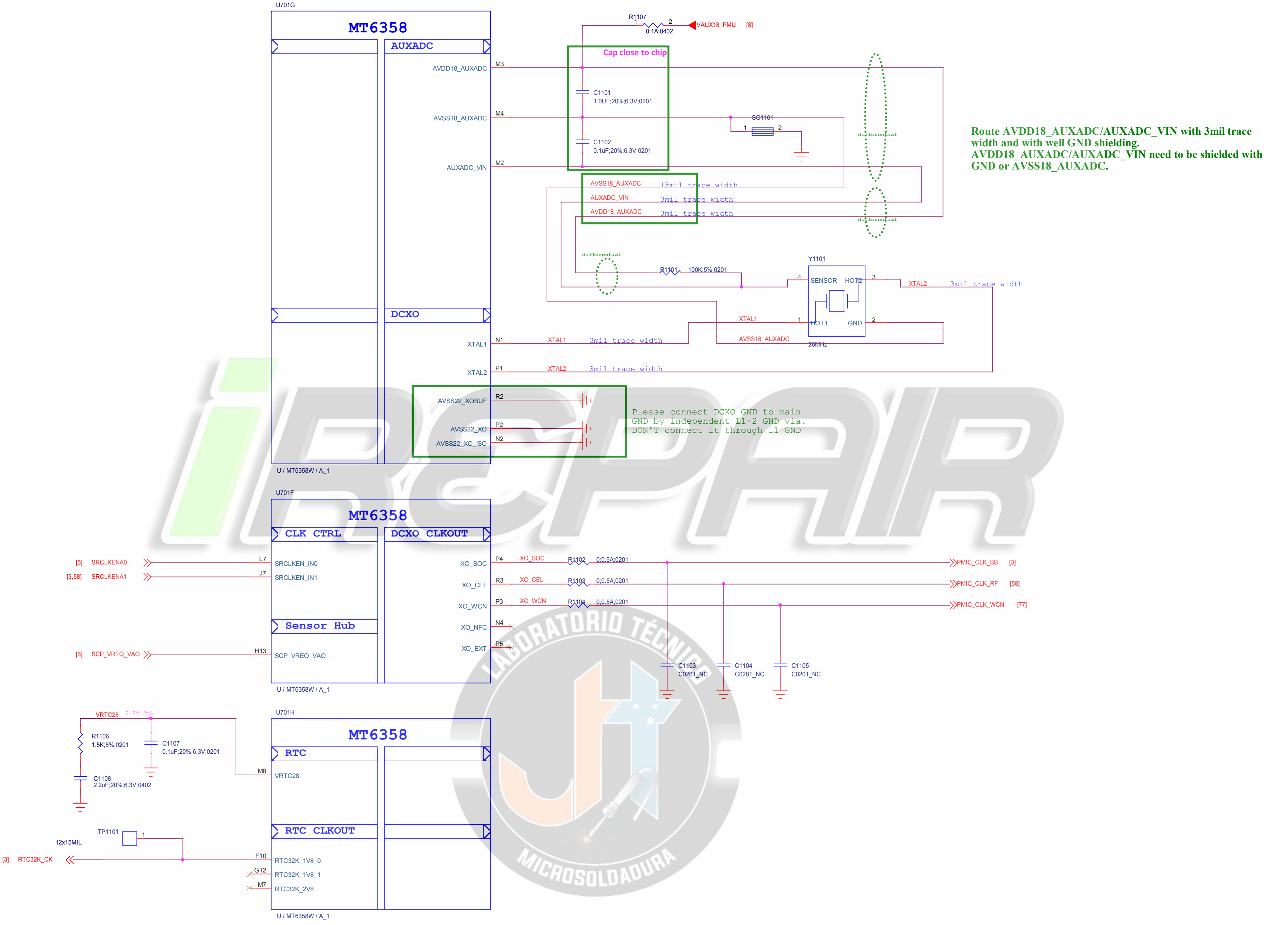

Title	11_POWER_MT6358_Clock	REV: V10	
DOCUMENT NO.:	Design Name	Size	C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Friday, May 10, 2019	Sheet 13	of 65

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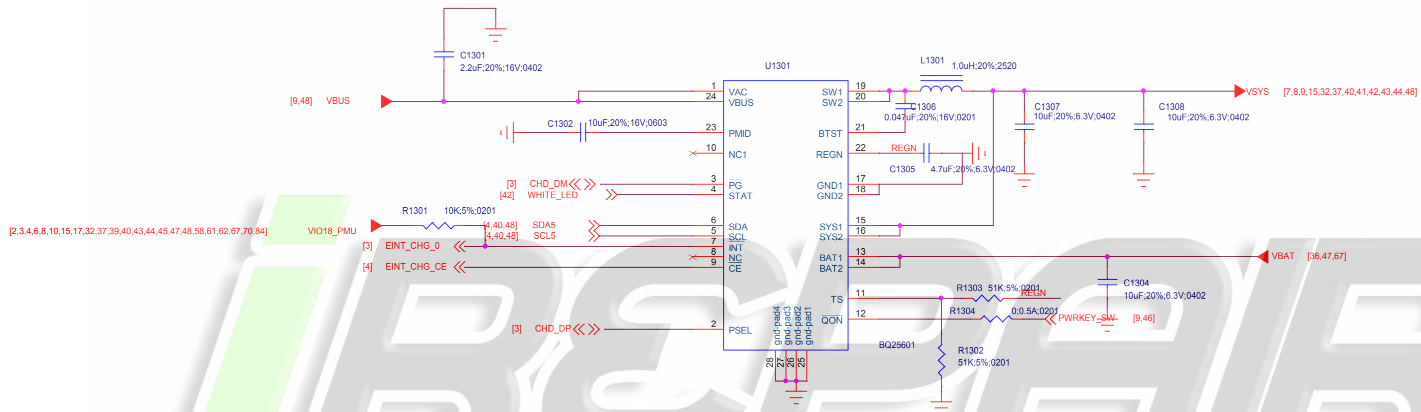
Title	12_POWER_MT6370-General		REV: V10
DOCUMENT NO.:	Design Name		Size C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Friday, May 10, 2019	Sheet 14 of 65	



Title	13_POWER_MT6370-Charger + PP		REV: V10
DOCUMENT NO.:	Design Name		Size C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Friday, May 10, 2019	Sheet	15 of 65

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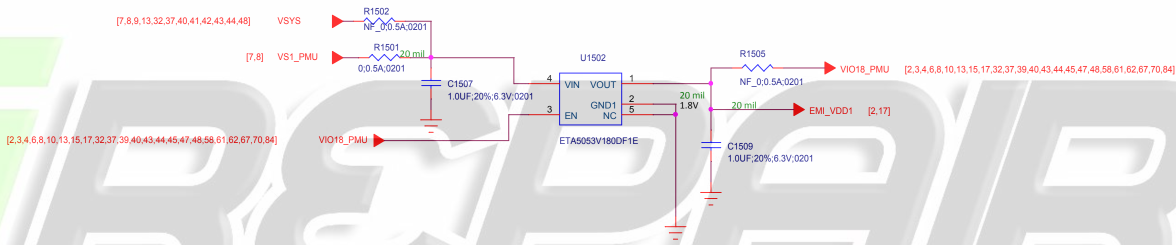
Schematic design notice of "27_POWER_SubPMIC-HV powers" page.

Note 27-1: It is recommended to reserve 0-ohm and cap. for BOM fine tune to minimize RF de-sense.

Note 27-2: It is recommended to reserve 0-ohm for BOM fine tune to minimize RF de-sense.

Title	14_POWER_MT6370-HV powers		REV: V10
DOCUMENT NO.:	Design Name		Size C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Friday, May 10, 2019	Sheet 16 of 65	

LDO for EMI_VDD1 of LPDDR4 VDD1



Title	15_POWER_ThirdParty_Powers		REV: V10
DOCUMENT NO.:	Design Name		Size C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
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Title	16_BB_ POWER_PDN		REV: V10
DOCUMENT NO.:	Design Name		Size C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
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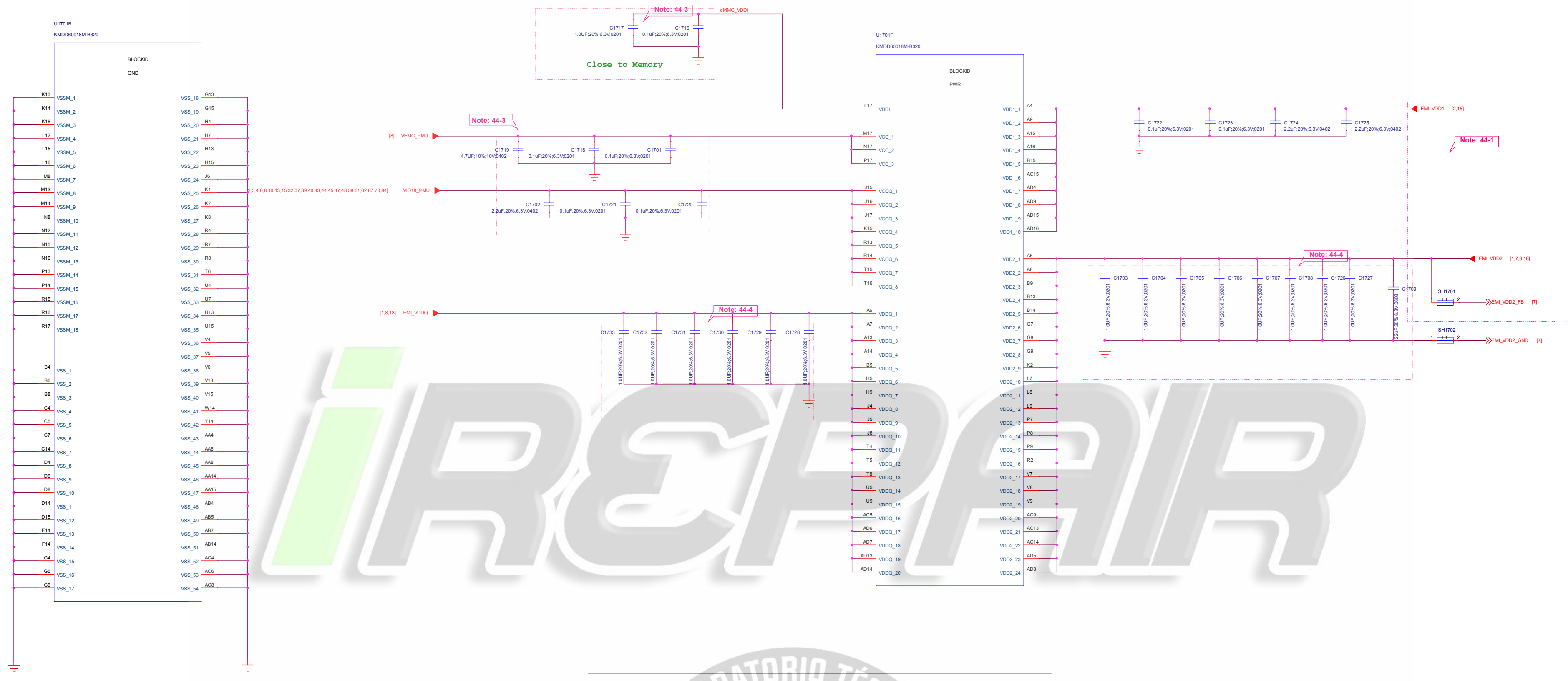

Schematic design notice of "44_Memory_eMMC_LPDDR4X"

Note 44-1: Please refer to power supply related page select VDRAM 2 / VDRAM1 output voltage properly for LPDDR4X

Note 44-2: DRAM ZQx resistor = 240ohm (1%) that must be connected to VDDQ,

Note 44-3: Please refer to eMCP vendor's datasheet or MTK common design notice to get the recommendation bypass cap. value for VCC/VCCQ/VDDI power domains of eMMC.

Note 44-4: VDD2 VDDQ decoupling cap: closed to DRAM ball.
For other cap for PMIC [>10uF, at PMIC page]:
please also refer to MMD and layout guide for placement.

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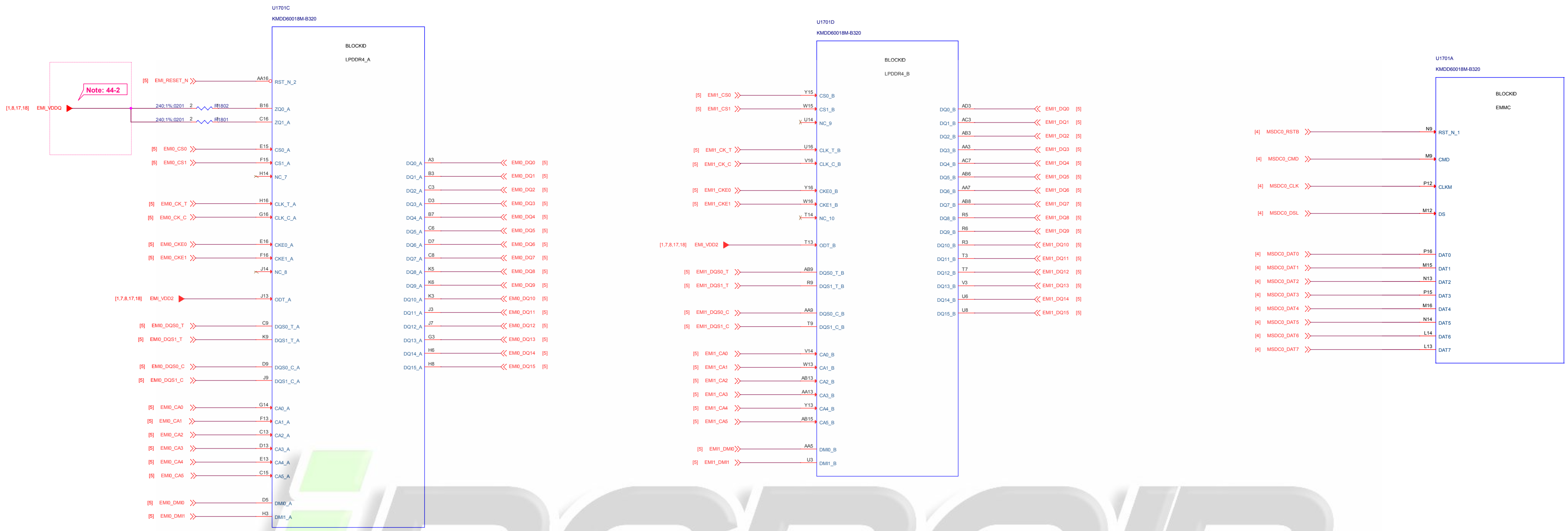
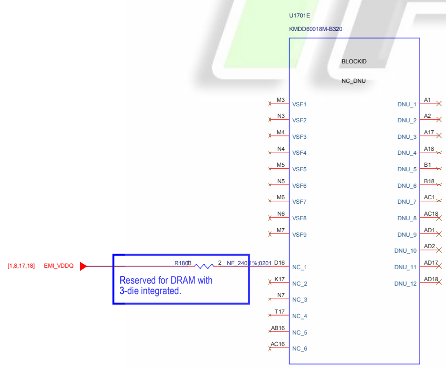
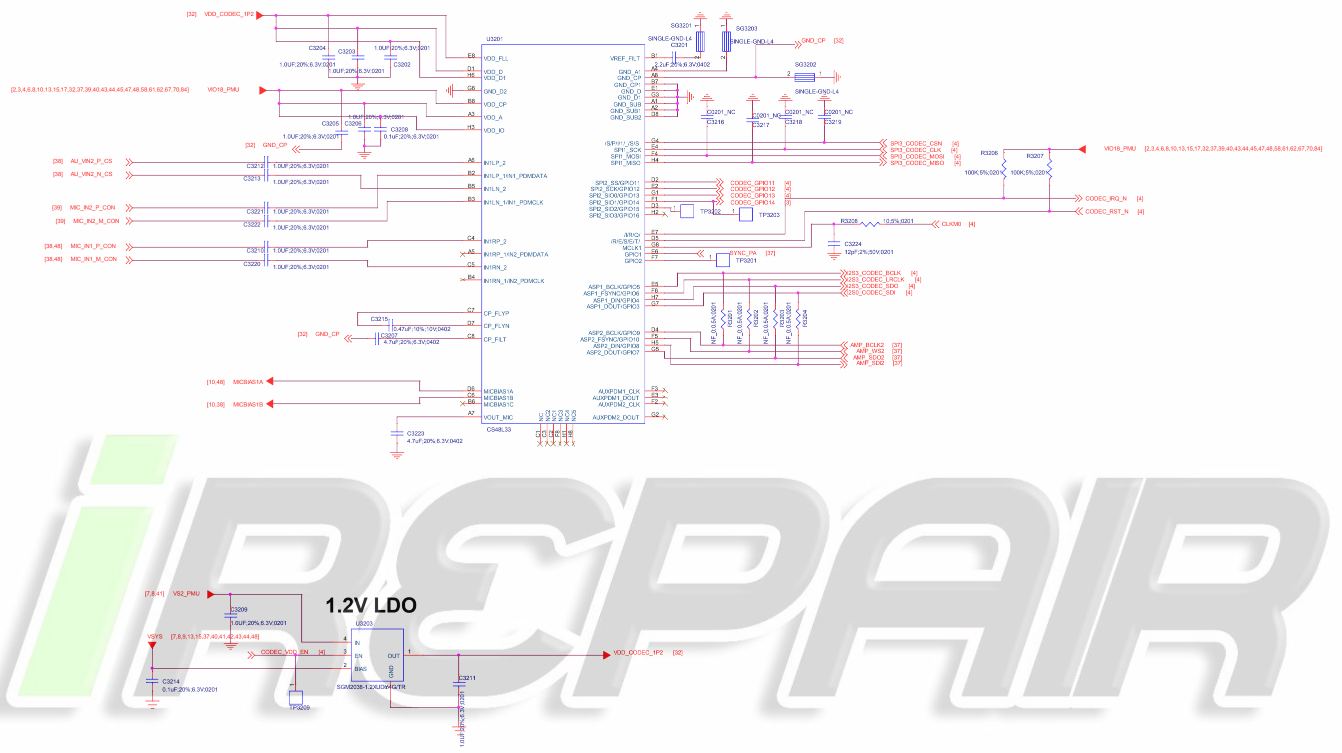
Title	17_Memory_LPDDR4_POWER	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Friday, May 10, 2019	Sheet 19 of 65	





Schematic design notice of "44_Memory_eMMC_LPDDR4X"	
Note 44-1:	Please refer to power supply related page select VDRAM 2 / VDRAM1 output voltage properly for LPDDR4X
Note 44-2:	DRAM ZQx resistor = 240ohm (1%) that must be connected to VDDQ,
Note 44-3:	Please refer to eMCP vendor's datasheet or MTK common design notice to get the recommendation bypass cap. value for VCC/VCCQ/VDDI power domains of eMMC.
Note 44-4:	VDD2 VDDQ decoupling cap: closed to DRAM ball. For other cap for PMIC [>10uF, at PMIC page]: please also refer to MMD and layout guide for placement.

Title	18_Memory_eMMC_LPDDR4	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Friday, May 10, 2019	Sheet	20 of 65



Title	32.CODEC_Cirrus		REV: V10
DOCUMENT NO.:	98860_1_12_201905111250		Size D
DEPARTMENT:	DEPARTMENT	DESIGNER:	DESIGNER
WINGTECH			
Date:	Friday, May 10, 2019	Sheet	21 of 65

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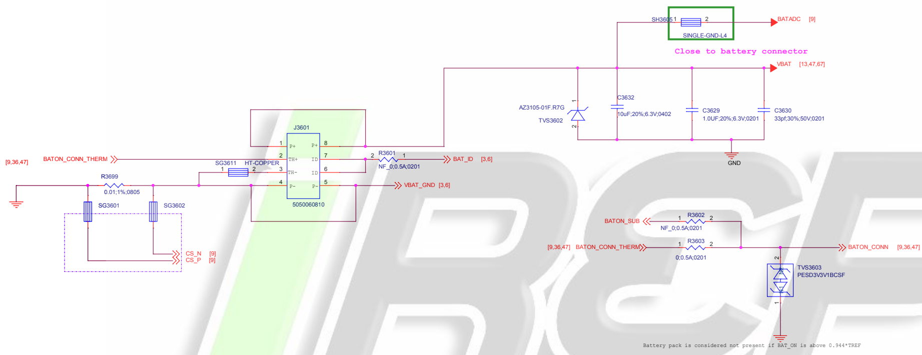
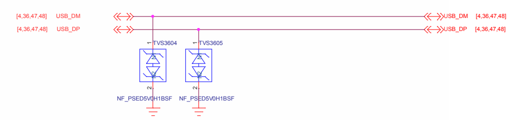
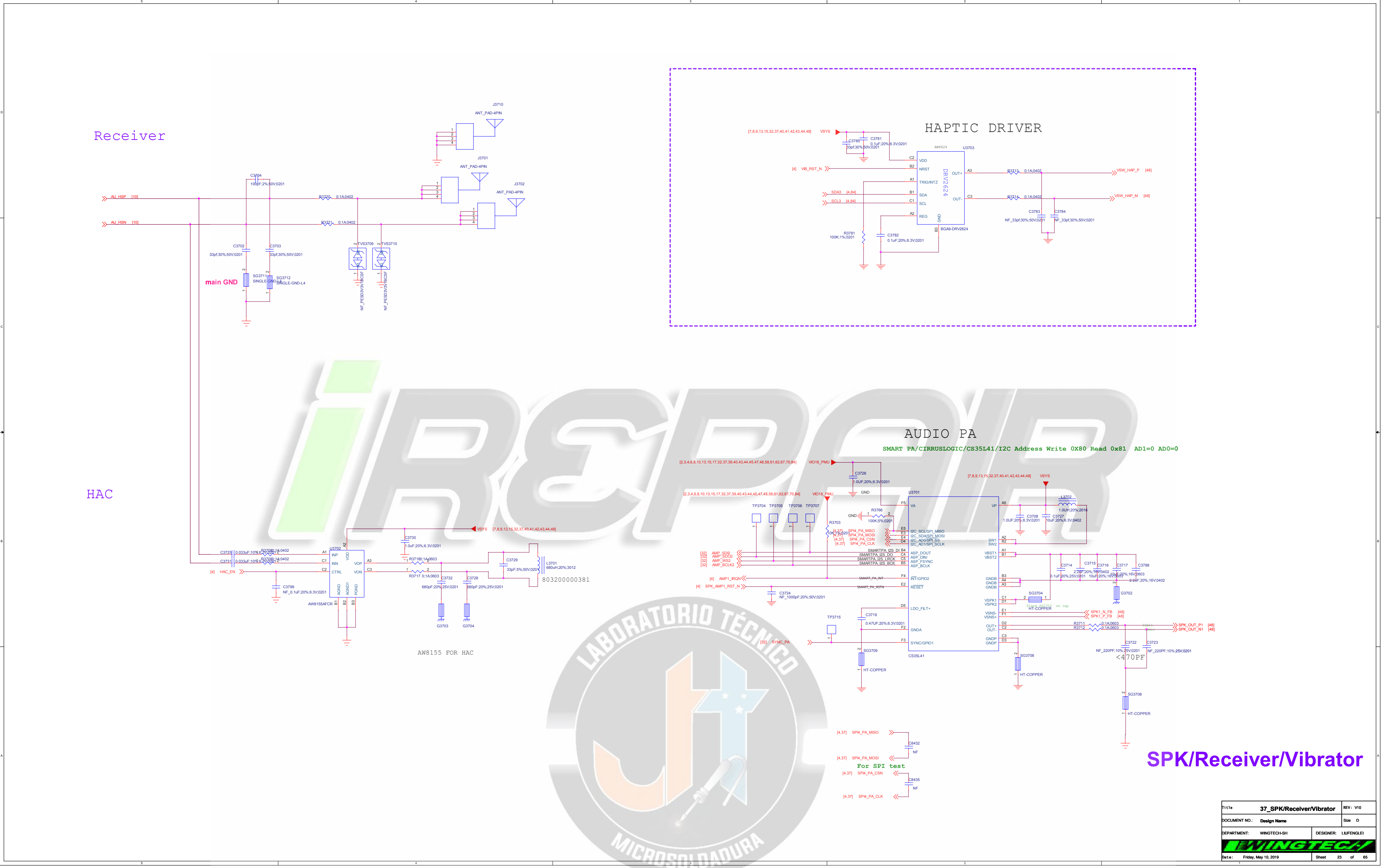
BATTERY CONNECTOR





Title	36_Battery/USB IF	REV: V10
DOCUMENT NO.:	Design Name	Size D
DEPARTMENT: WINGTECH-SH	DESIGNER: LIUFENGLEI	
WINGTECH		
Date: Saturday, May 11, 2019	Sheet 22 of 65	



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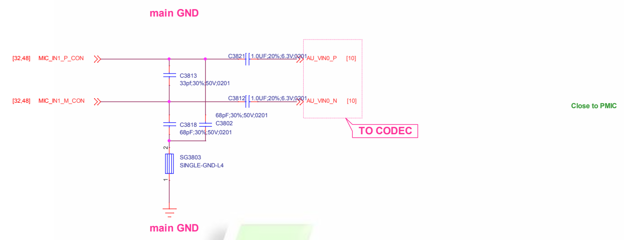
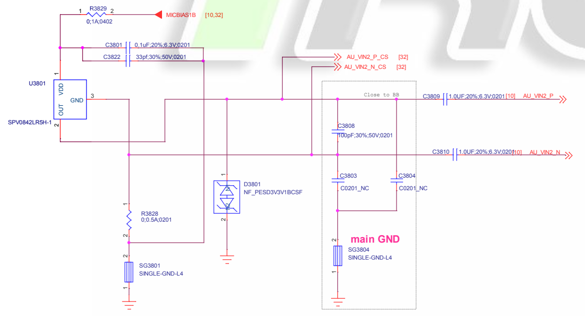
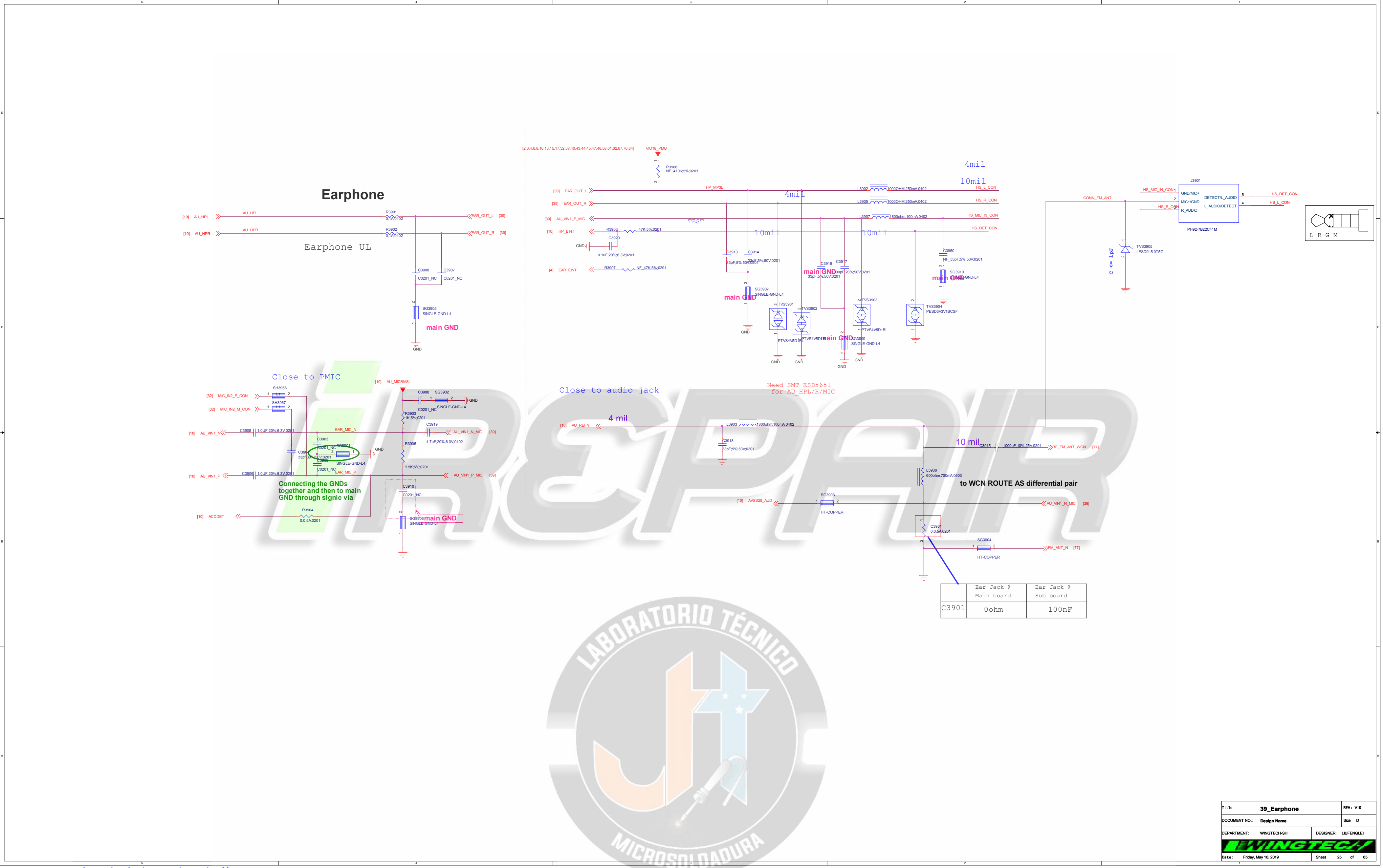
MAIN MIC



TOP MIC



Title	38_MIC		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Friday, May 10, 2019	Sheet	24 of 65



Schematic design notice of "62_PERI_AUDIO_IO" page.

Note 62-1: Part # of BEAD6202, BEAD6203, BEAD6204 and BEAD6205 needs changed to "BLM18BD102SN1" for high THD performance (-90dB) but this BOM change will results in FM RSSI 10dB degraded .

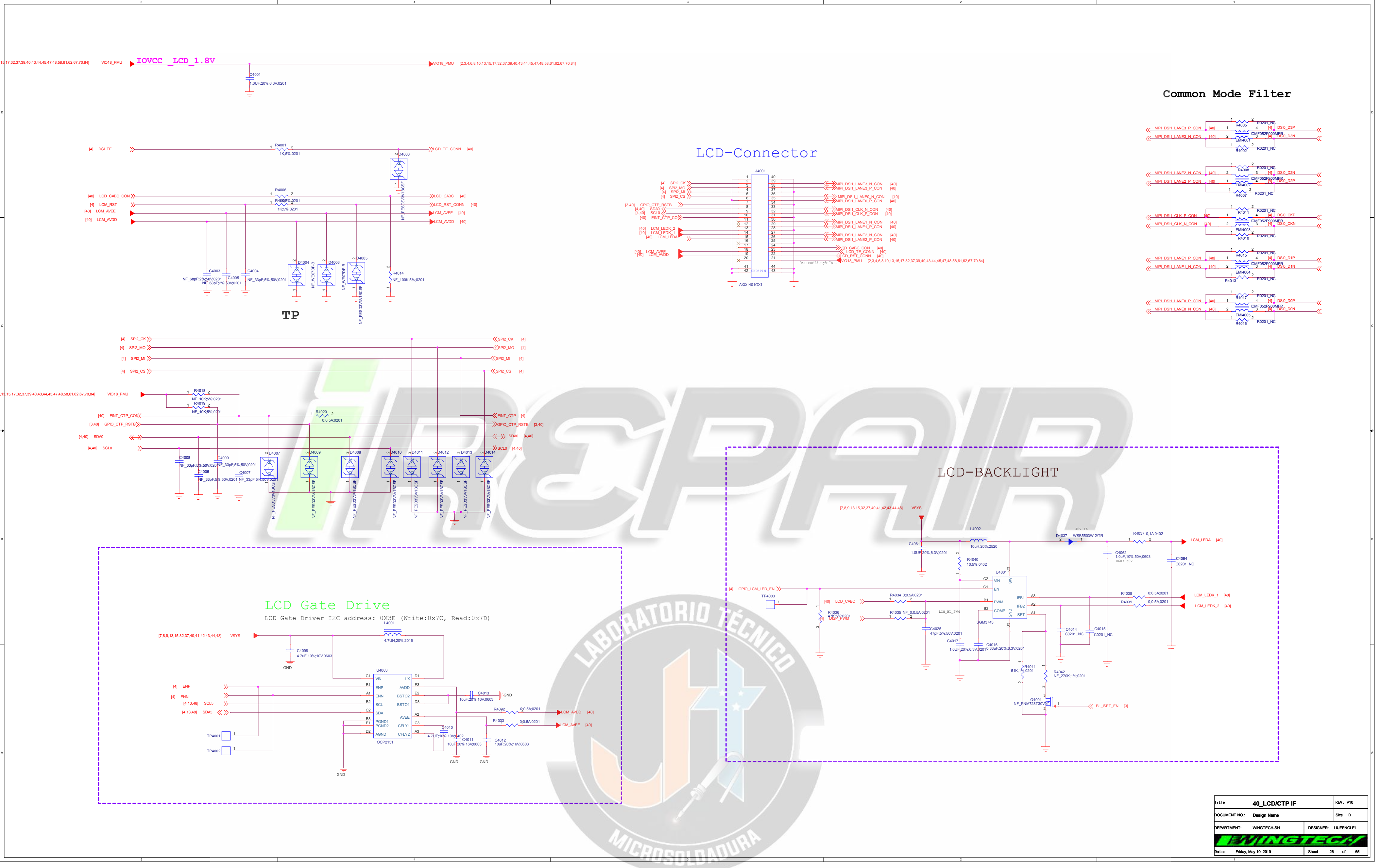
Note 62-2: Reserved Cap for CS/RS test, please double check multi-key function when used

Note 62-3:

	Earphone Jack @ Main board	Earphone Jack @ Sub board
R6513	0 ohm	100nF

Note 62-4: Please Select ACC Mode for Operator Project to Pass Electrical MOS Test;
More Information refer to Audio/Speech Design Notice

Note 62-5: Please select R6231 with 0402 size


IOVCC _LCD_1.8V
Common Mode Filter
LCD-Connector
TP
LCD-BACKLIGHT
LCD Gate Drive
LCD Gate Driver I2C address: 0X3E (Write:0x7C, Read:0x7D)
OCP2131
SGM3743
AXQ1401GX1
Title 40_LCD/CTP IF
REV: V10
DOCUMENT NO.: Design Name
Size D
DEPARTMENT: WINGTECH-SH
DESIGNER: LIUFENGLEI
WINGTECH
Date: Friday, May 10, 2019
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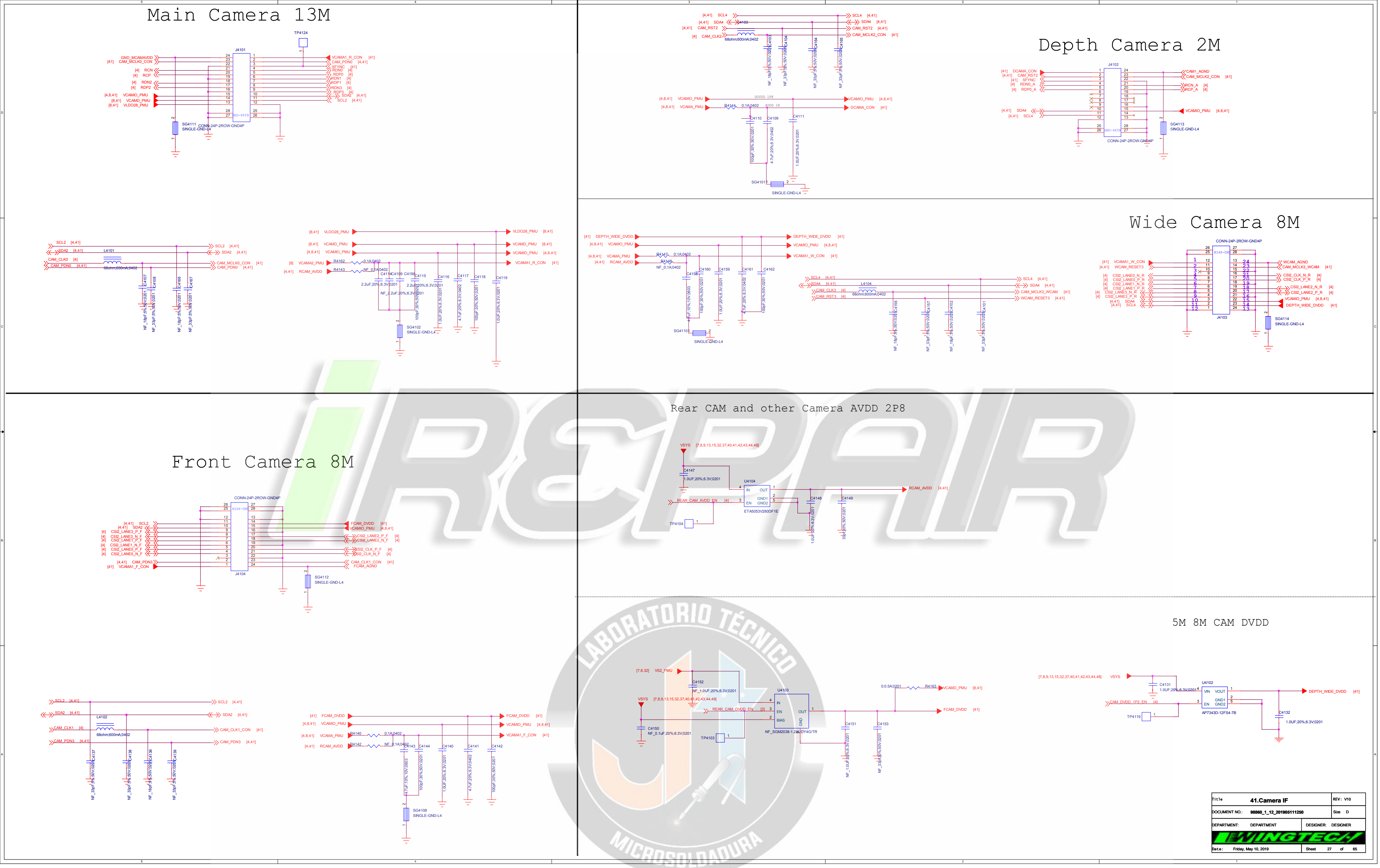
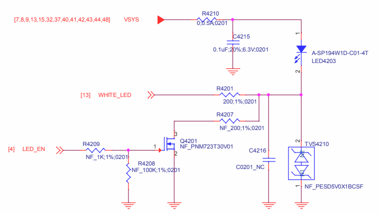
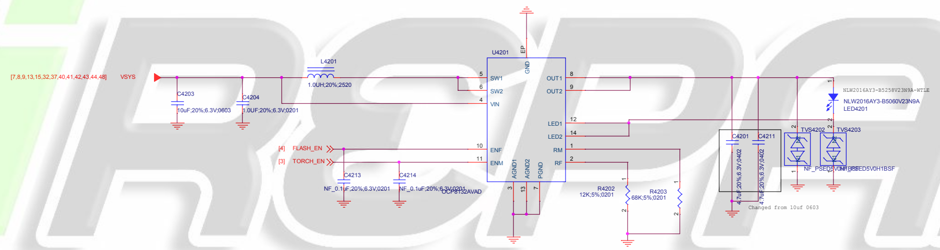

Main Camera 13M
Depth Camera 2M
Wide Camera 8M
Front Camera 8M
Rear CAM and other Camera AVDD 2P8
5M 8M CAM DVDD
41.Camera IF
REV: V10
Friday, May 10, 2019
Sheet 27 of 65
WINGTECH



Input and output cap voltage confirm ?



Title	42_Flash/RGB		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Friday, May 10, 2019	Sheet	28 of 65

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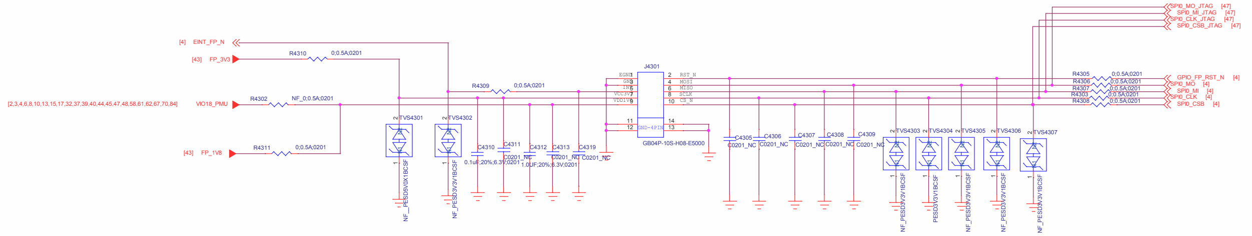
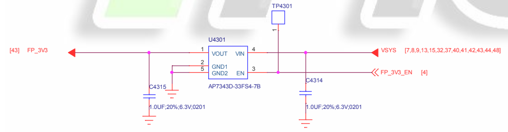
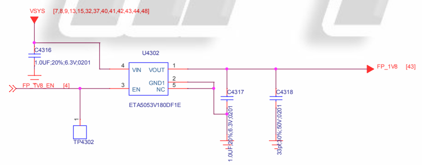
MICROSOLDADURA

Fingerprint

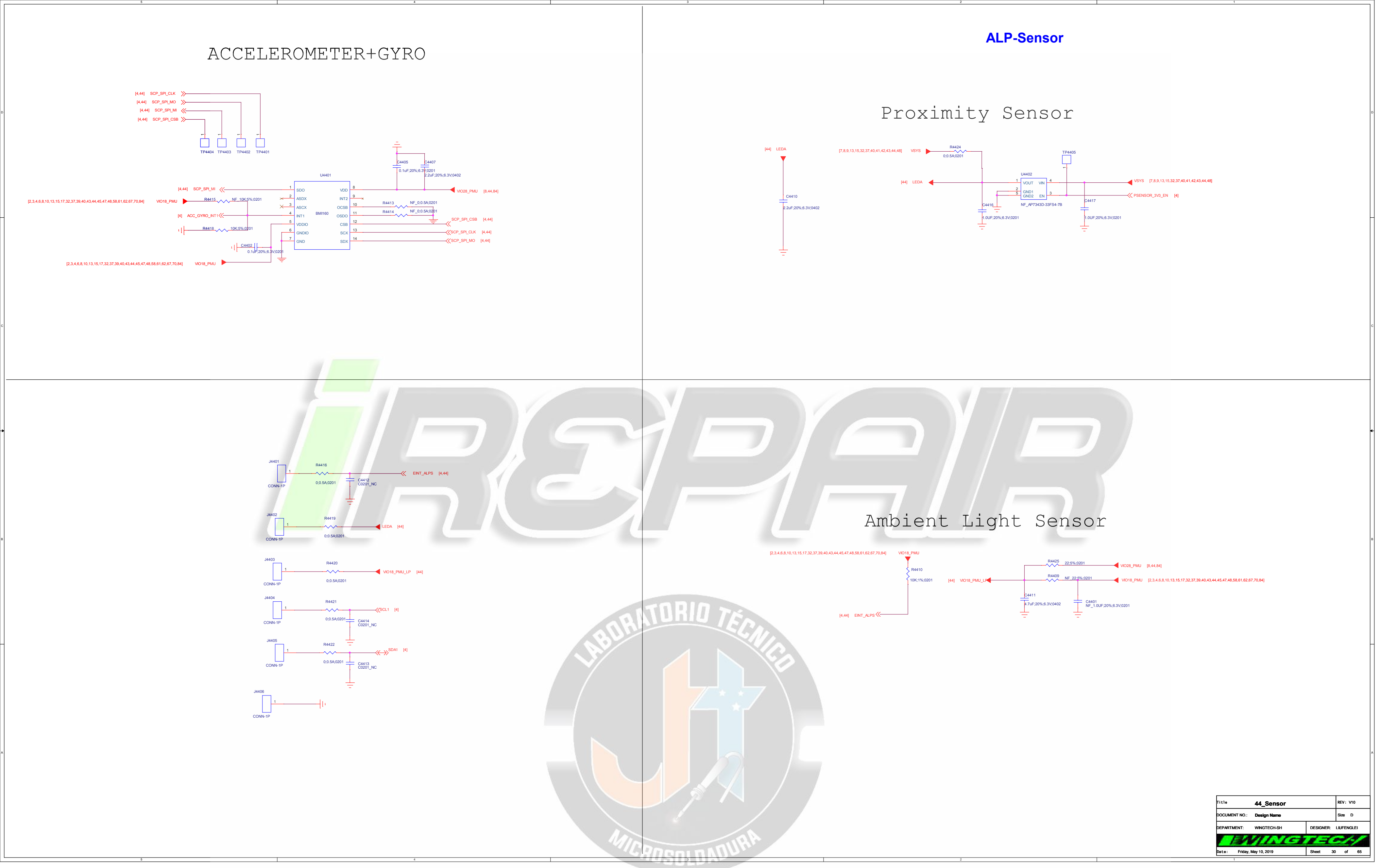


Fingerprint





Title	43.Fingerprint	REV: V10	
DOCUMENT NO.:	98860_1_12_201905111250	Size D	
DEPARTMENT:	Wingtech-SZ	DESIGNER:	DESIGNER
Date:	Friday, May 10, 2019	Sheet 29 of 65	


ACCELEROMETER+GYRO
ALP-Sensor
Proximity Sensor
Ambient Light Sensor
SCP_SPI_CLK
SCP_SPI_MO
SCP_SPI_MI
SCP_SPI_CSB
TP4404
TP4403
TP4402
TP4401
U4401
BMI160
SDO
ASDX
ASCX
INT1
VDDIO
GNDIO
GND
VDD
INT2
OCSB
OSDO
CSB
SCX
SDX
ACC_GYRO_INT1
VIO18_PMU
VIO28_PMU
R4413
R4414
R4415
R4418
C4402
C4405
C4407
LEDA
VSYS
R4424
TP4405
U4402
VOUT
VIN
GND1
GND2
EN
NF_AP7343D-33FS4-7B
PSENSOR_3V3_EN
C4410
C4416
C4417
J4401
J4402
J4403
J4404
J4405
J4406
CONN-1P
R4416
R4419
R4420
R4421
R4422
C4412
C4413
C4414
EINT_ALPS
VIO18_PMU_LP
SCL1
SDA1
R4410
R4425
R4409
C4411
C4401
Title 44_Sensor
REV: V10
DOCUMENT NO.: Design Name
Size D
DEPARTMENT: WINGTECH-SH
DESIGNER: LIUFENGLEI
WINGTECH
Date: Friday, May 10, 2019
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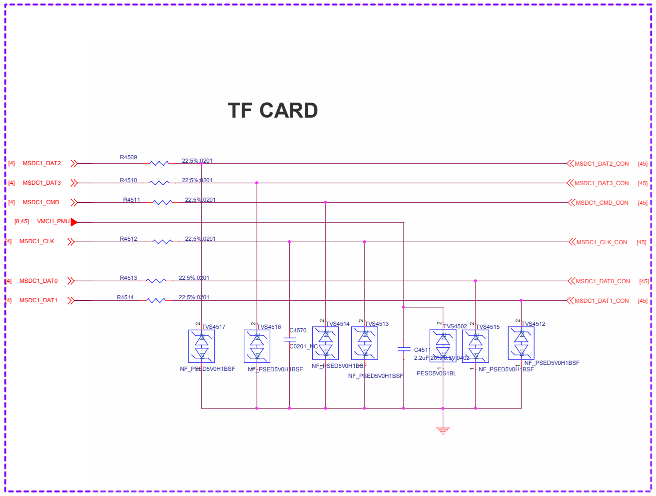
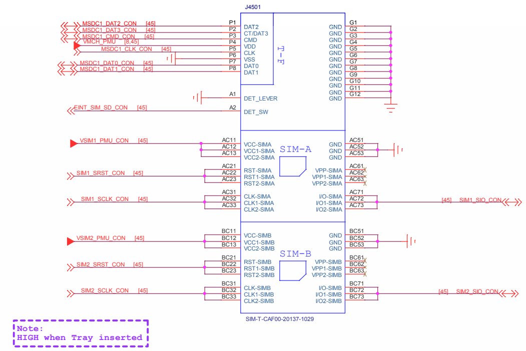
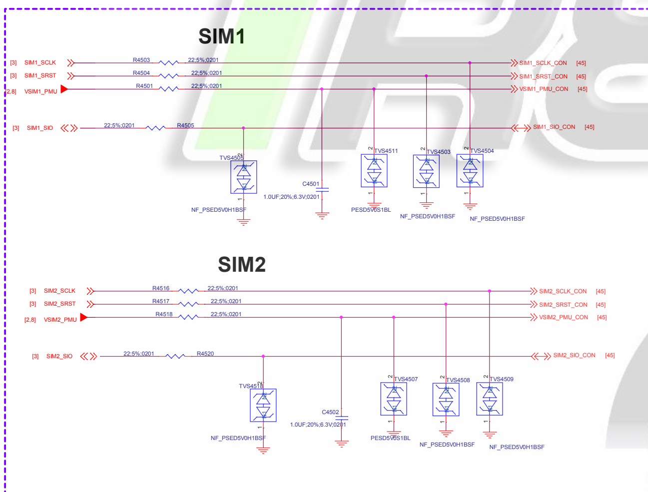
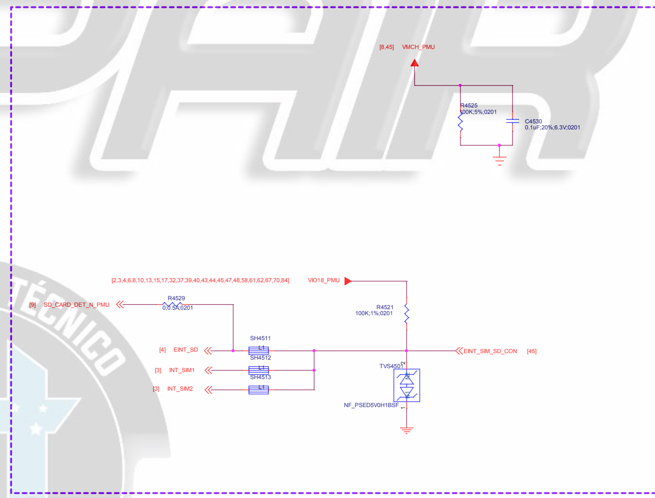




SIM&SD Connector





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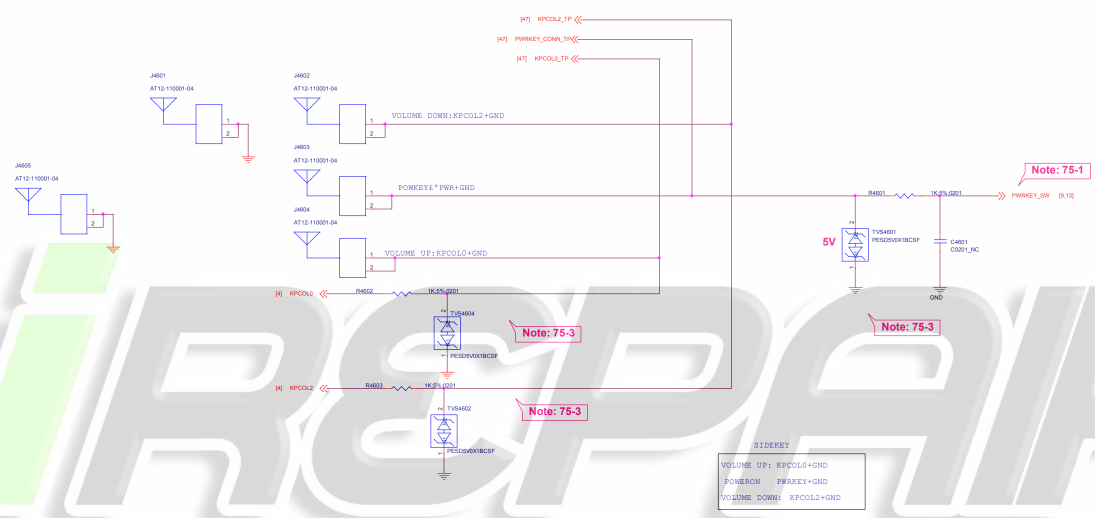
MICROSOLDADURA

Title	45_SIM/TF IF	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Friday, May 10, 2019	Sheet 31 of 65	

Power Key / Key Pad

DO NOT put pull-up resistor on PWRKEY



Note 75-1:DO NOT put pull-up resistor on PWRKEY
Note 75-2: Volume Up:KPCOL0+GND
Volume Down:KPCOL2+GND

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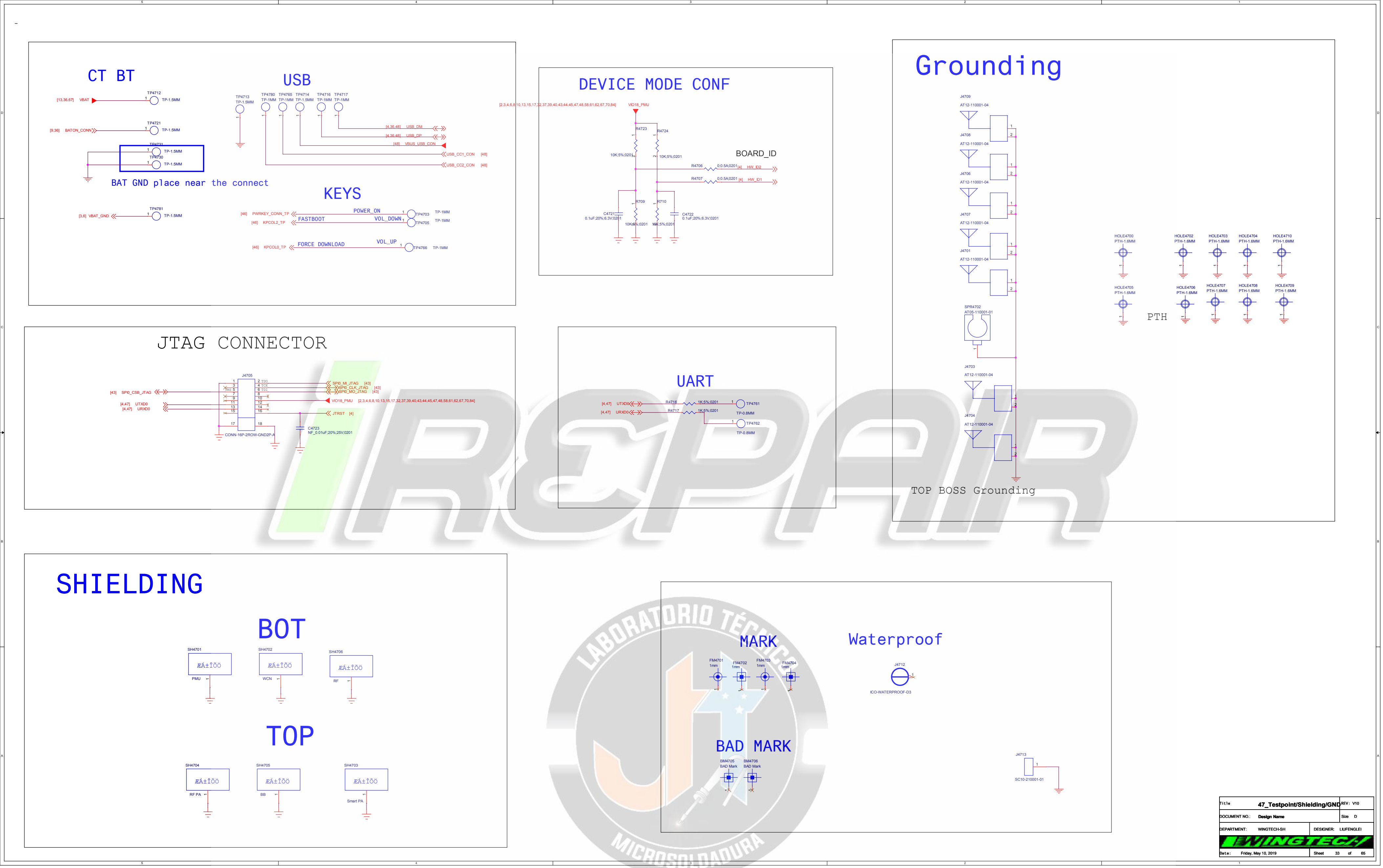
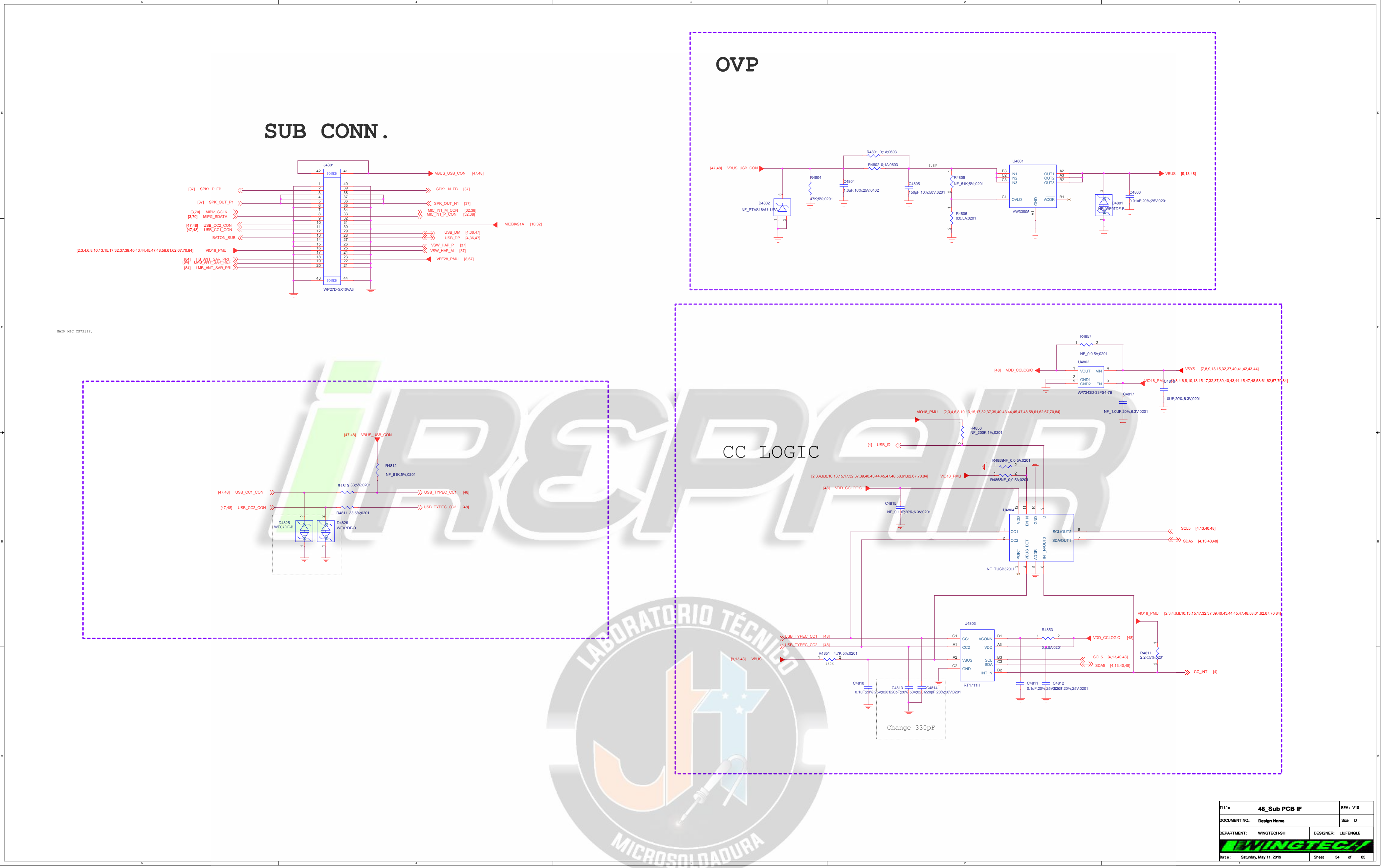
MICROSOLDADURA

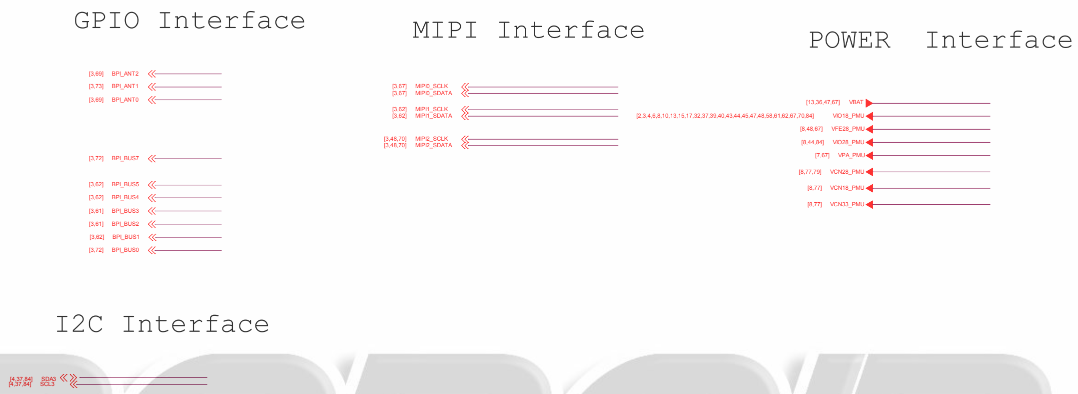
Title	46_Sidekey	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Friday, May 10, 2019	Sheet	32 of 65



CT BT
USB
KEYS
TP4712 TP-1.5MM VBAT
TP4721 TP-1.5MM BATON_CONN
TP4731 TP-1.5MM
TP4730 TP-1.5MM
BAT GND place near the connect
TP4781 TP-1.5MM VBAT_GND
TP4713 TP-1.5MM
TP4780 TP-1MM
TP4765 TP-1MM
TP4714 TP-1.5MM
TP4716 TP-1MM
TP4717 TP-1MM
USB_DM
USB_DP
VBUS_USB_CON
USB_CC1_CON
USB_CC2_CON
PWRKEY_CONN_TP POWER_ON TP4703 TP-1MM
KPCOL2_TP FASTBOOT VOL_DOWN TP4705 TP-1MM
KPCOL0_TP FORCE DOWNLOAD VOL_UP TP4766 TP-1MM
DEVICE MODE CONF
VIO18_PMU
R4723
R4724
10K;5%;0201
BOARD_ID
R4706 0;0.5A;0201 HW_ID2
R4707 0;0.5A;0201 HW_ID1
R709
R710
C4721 0.1uF;20%;6.3V;0201
C4722 0.1uF;20%;6.3V;0201
Grounding
J4709 AT12-110001-04
J4708 AT12-110001-04
J4706 AT12-110001-04
J4707 AT12-110001-04
J4701 AT12-110001-04
SPR4702 AT05-110001-01
J4703 AT12-110001-04
J4704 AT12-110001-04
TOP BOSS Grounding
HOLE4700 PTH-1.6MM
HOLE4702 PTH-1.6MM
HOLE4703 PTH-1.6MM
HOLE4704 PTH-1.6MM
HOLE4710 PTH-1.6MM
HOLE4705 PTH-1.6MM
HOLE4706 PTH-1.6MM
HOLE4707 PTH-1.6MM
HOLE4708 PTH-1.6MM
HOLE4709 PTH-1.6MM
PTH
JTAG CONNECTOR
J4705
SPI0_CSB_JTAG
UTXD0
URXD0
SPI0_MI_JTAG
SPI0_CLK_JTAG
SPI0_MO_JTAG
VIO18_PMU
JTRST
CONN-16P-2ROW-GND2P-A
C4723 NF_0.01uF;20%;25V;0201
UART
UTXD0 R4716 1K;5%;0201 TP4761 TP-0.8MM
URXD0 R4717 1K;5%;0201 TP4762 TP-0.8MM
SHIELDING
BOT
SH4701 PMU
SH4702 WCN
SH4706 RF
TOP
SH4704 RF PA
SH4705 BB
SH4703 Smart PA
MARK
FM4701 1mm
FM4702 1mm
FM4703 1mm
FM4704 1mm
BAD MARK
BM4705 BAD Mark
BM4706 BAD Mark
Waterproof
J4712 ICO-WATERPROOF-D3
J4713 SC10-210001-01
Title 47_Testpoint/Shielding/GND
REV: V10
DOCUMENT NO.: Design Name
Size D
DEPARTMENT: WINGTECH-SH
DESIGNER: LIUFENGLEI
WINGTECH
Date: Friday, May 10, 2019
Sheet 33 of 65



OVP
SUB CONN.
CC LOGIC
MAIN MIC CS7331P.
J4801
WP27D-SX40VA3
VBUS_USB_CON
SPK1_P_FB
SPK1_N_FB
SPK_OUT_P1
SPK_OUT_N1
MIPI2_SCLK
MIPI2_SDATA
MIC_IN1_M_CON
MIC_IN1_P_CON
USB_CC2_CON
USB_CC1_CON
MICBIAS1A
BATON_SUB
USB_DM
USB_DP
VSW_HAP_P
VSW_HAP_M
VIO18_PMU
VFE28_PMU
U4801
AW33905
R4801 0.1A;0603
R4802 0.1A;0603
R4804
R4805
R4806
C4804
C4805
C4806
D4801
D4802
VBUS
U4802
AP7343D-33FS4-7B
R4857
VSYS
VDD_CCLOGIC
C4817
C4818
R4856
USB_ID
R4859
R4858
C4815
U4804
NF_TUSB320LI
SCL5
SDA5
U4803
RT1711H
R4853
R4851
R4817
CC_INT
C4810
C4811
C4812
C4813
C4814
Change 330pF
USB_TYPEC_CC1
USB_TYPEC_CC2
R4810
R4811
R4812
D4825
D4826
Title 48_Sub PCB IF
REV: V10
DOCUMENT NO.: Design Name
Size D
DEPARTMENT: WINGTECH-SH
DESIGNER: LIUFENGLEI
WINGTECH
Date: Saturday, May 11, 2019
Sheet 34 of 65



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MICROSOLDADURA

Title	Page Name = 56_RF Interface	REV: V10	
DOCUMENT NO.:	Design Name = 98860_1_12_201905111250	Size D	
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
WINGTECH			
Date:	Page Modify Date = Friday, May 10, 2019	Sheet 35 of 65	

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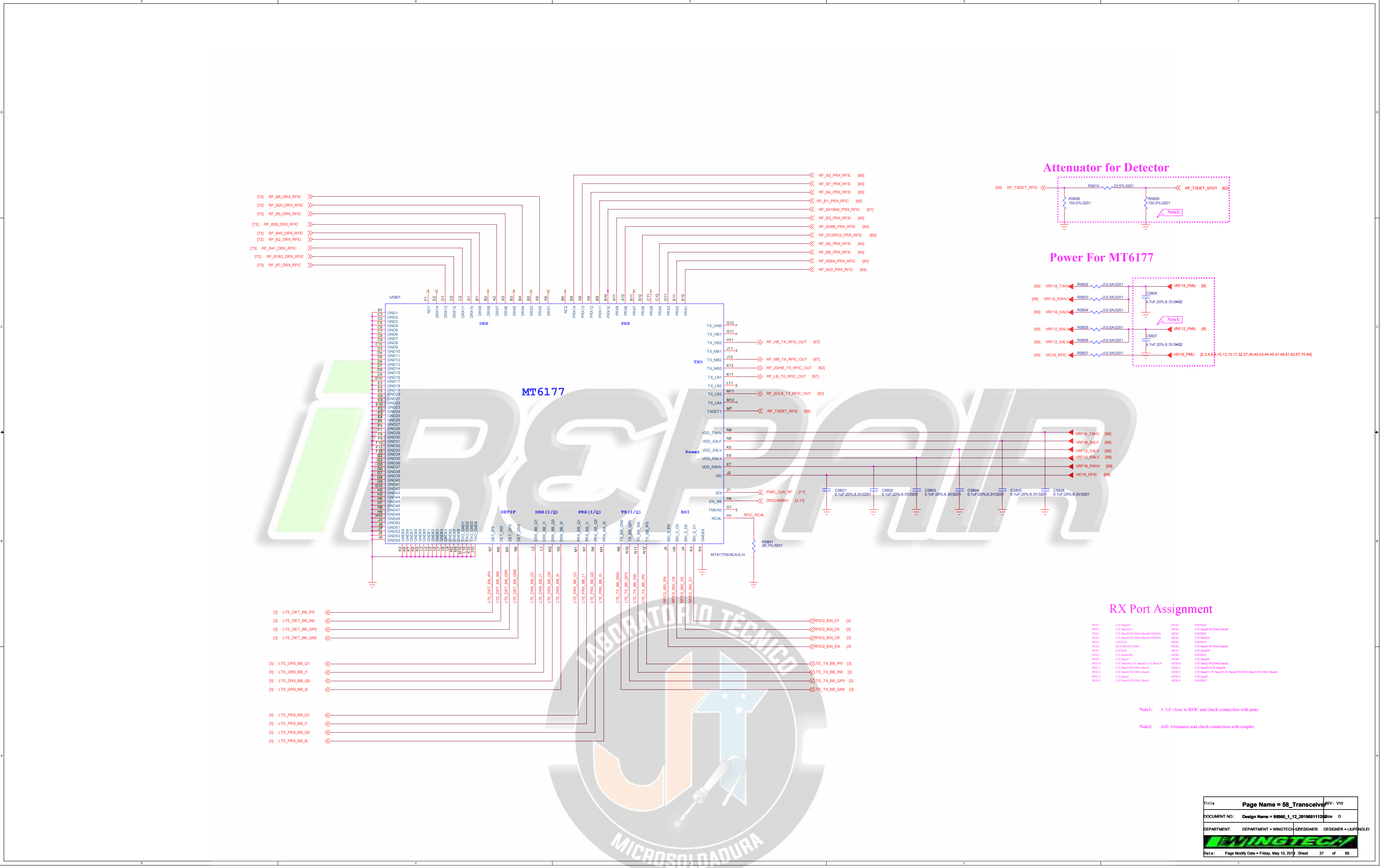
LABORATORIO TÉCNICO
MICROSOLDADURA

Title	<Title>		
Size A1	Document Number <Doc>		Rev 1.0
Date:	Friday, May 10, 2019	Sheet 36	of 65


MT6177
U5801
DRX
PRX
TXO
Power
DETIF
DRX(I/Q)
PRX(I/Q)
TX(I/Q)
BSI
MT6177W/ACA-E-H
Attenuator for Detector
Power For MT6177
RX Port Assignment
Note1: 4.7uF close to RFIC and check connection with pmic
Note2: 6dB Attenuator and check connection with coupler
Page Name = 58_Transceiver
REV: V10
Department = WINGTECH-SZ
Designer = LIUFENGLEI
Page Modify Date = Friday, May 10, 2019
Sheet 37 of 65

iREPAIR
LABORATORIO TÉCNICO
MICROSOLDADURA

Title	Reserved	REV:	V10
DOCUMENT NO.:	Design Name	Size	A1
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Friday, May 10, 2019	Sheet	38 of 65

D

C

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JH
MICROSOLDADURA

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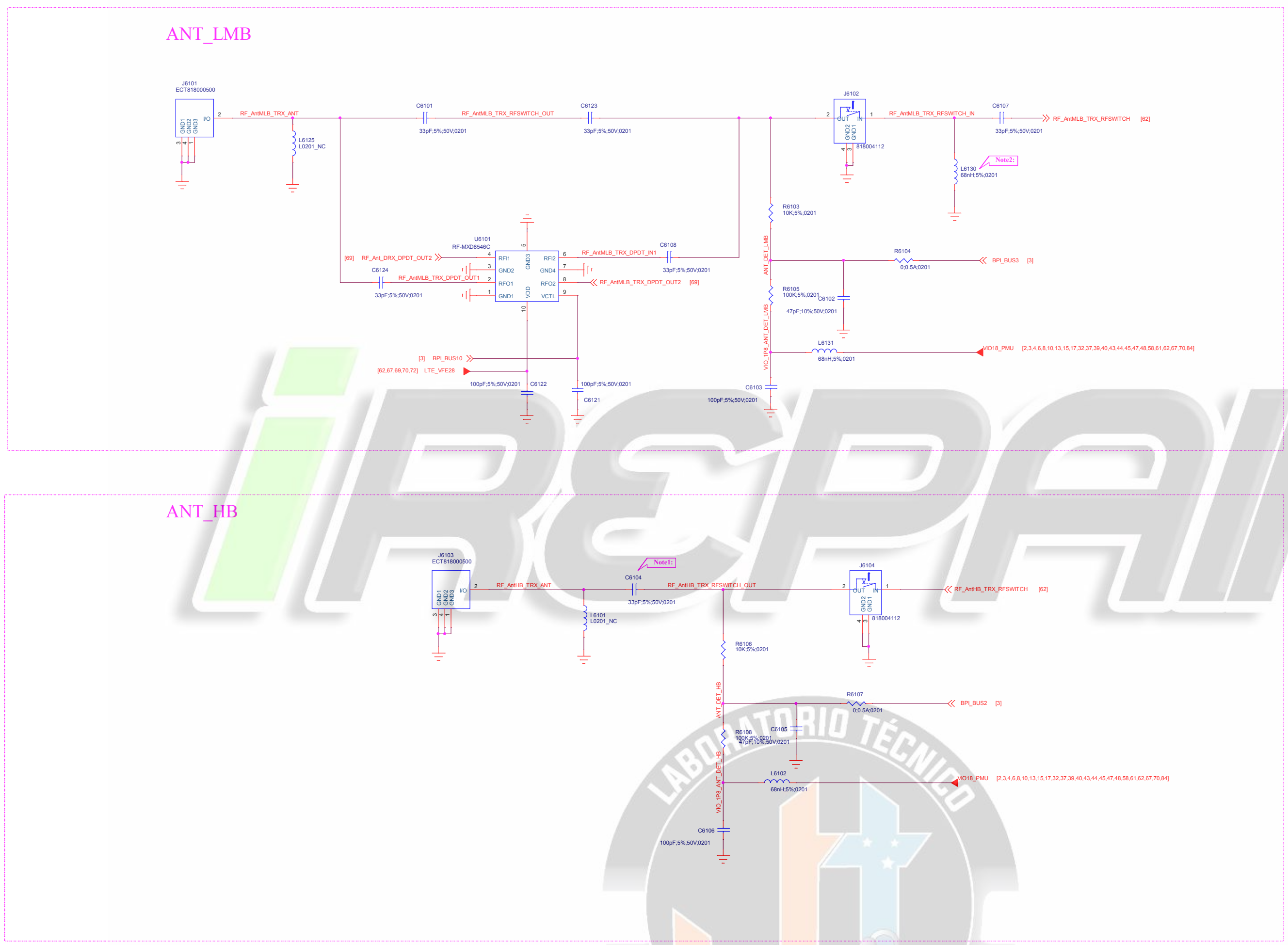
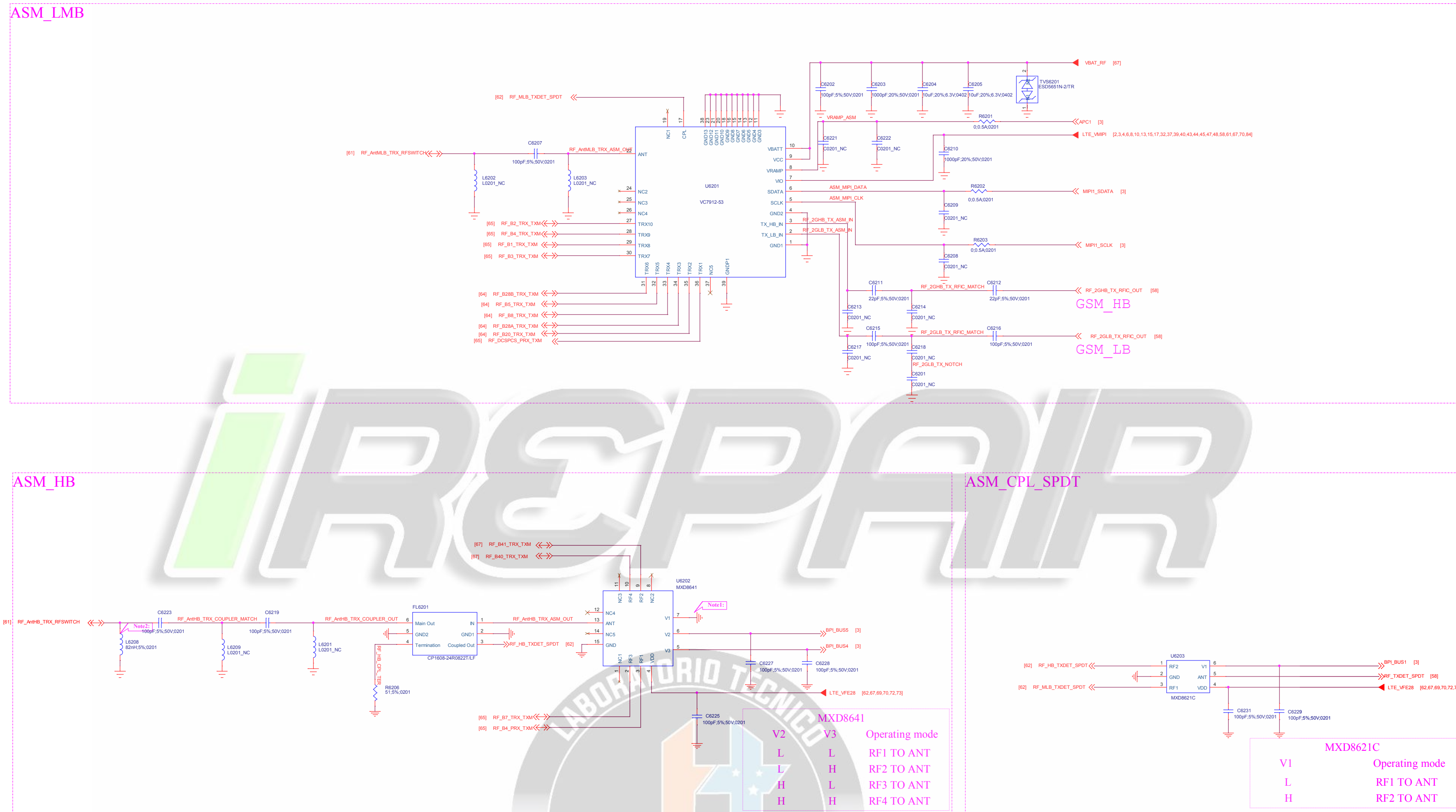
Title	Reserved	REV:	V10
DOCUMENT NO.:	Design Name	Size	A1
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Friday, May 10, 2019	Sheet	39 of 65



Note1: RF SWITCH DETECT,MUST BE CAP IF DPDT IS WITHOUT DC BLOCK

Note2: MUST BE IND FOR RF SWITCH DETECT

Title		REV: V10
DOCUMENT NO. Design Name = 98860_1_12_201905111250		Size D
DEPARTMENT: DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLEI	
WINGTECH		
Date: Page Modify Date = Friday, May 10, 2019	Sheet 40 of 65	



MXD8641

V2	V3	Operating mode
L	L	RF1 TO ANT
L	H	RF2 TO ANT
H	L	RF3 TO ANT
H	H	RF4 TO ANT

MXD8621C

V1	Operating mode
L	RF1 TO ANT
H	RF2 TO ANT

Note1: MXD8641 V1 CAN GROUD CONFIRMED WITH FAE

Note2: MUST BE IND FOR RF SWITCH DETECT

Title	Page Name = 62_Primary_ASM	REV: V10	
DOCUMENT NO.:	Design Name = 98860_1_12_20190511125	Size	D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
WINGTECH			
Date:	Page Modify Date = Friday, May 10, 2019	Sheet 41	of 65

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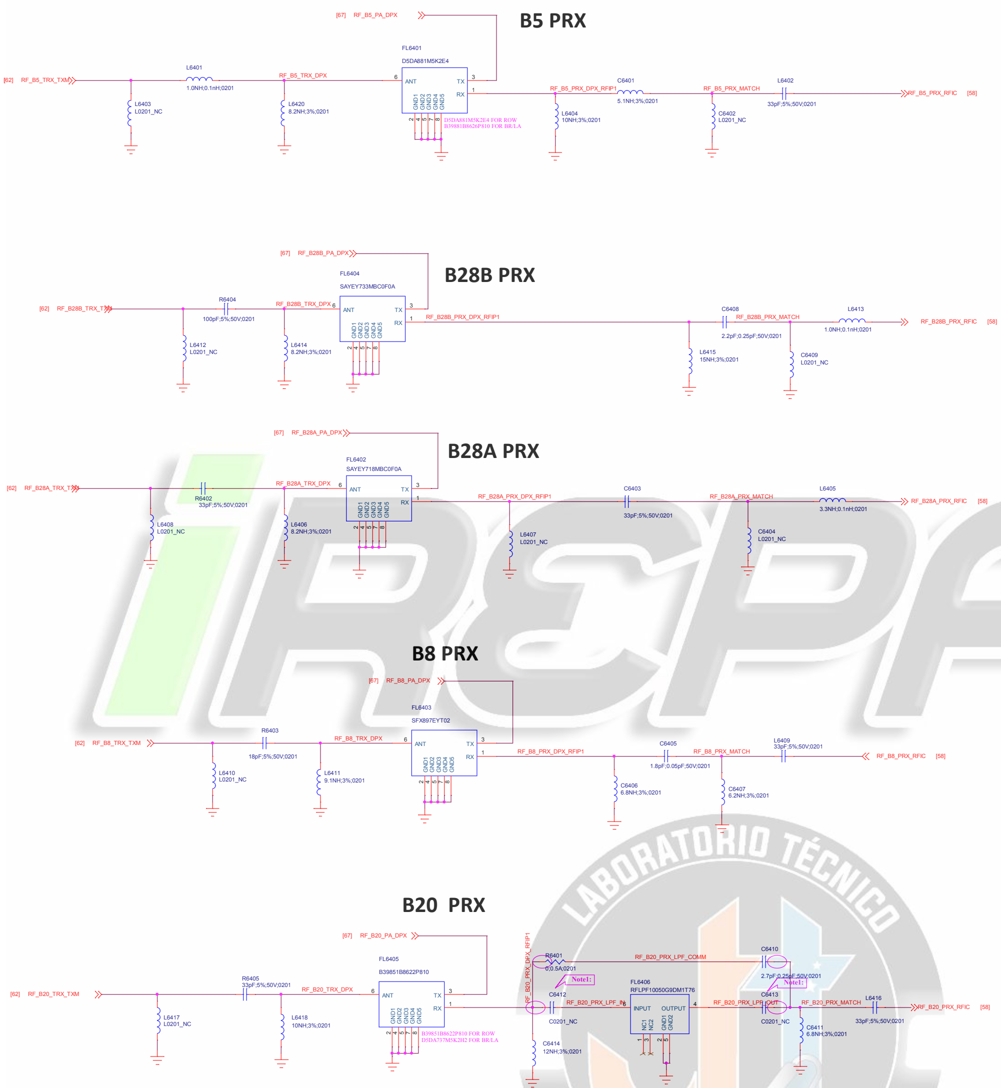
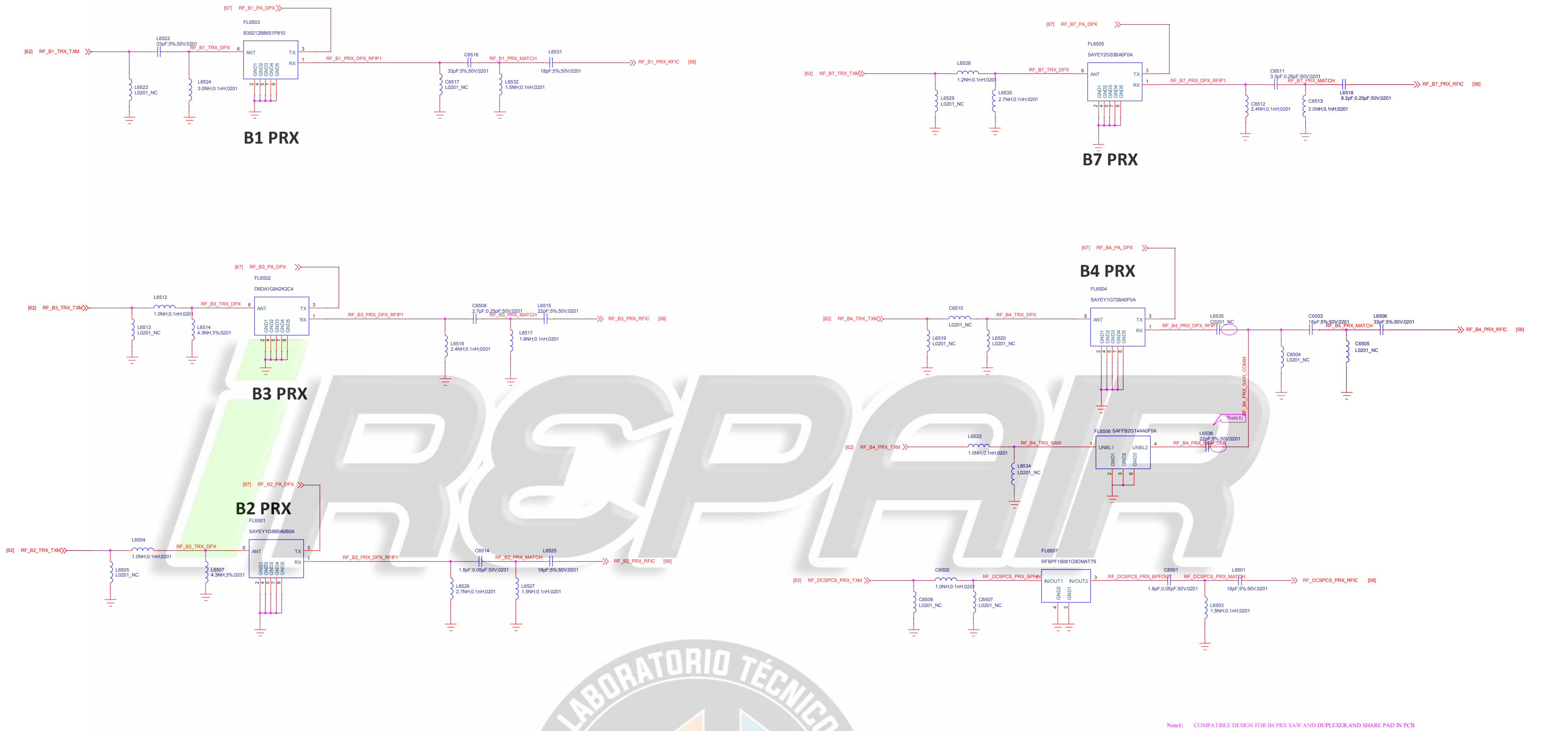
MICROSOLDADURA

Title	<Title>	
Size A1	Document Number <Doc>	Rev 1.0
Date: Friday, May 10, 2019	Sheet 42 of 65	



Note1: COMPATIBLE DESIGN FOR LPF AND SHARE PAD IN PCB

Title	Page Name = 64_TRX_LB	REV: V10
DOCUMENT NO.:	Design Name = 98860_1_12_20190511125	Size D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLEI
WINGTECH		
Date:	Page Modify Date = Friday, May 10, 2019	Sheet 43 of 65



Band Note:

	FL4704	FL4701	FL4707	FL4701	FL4702	FL4703	FL4702	FL4601	FL4602	FL4604	FL4603	U4802	U4805
China	B1	\	B3	\	G1900	\	B34/39	B5	B8	\	\	B40	B38/B41
CMCC	B1	B2	B3	\	G1900	B7	B34/39	B5	B8	\	\	B40	B38/B41
Asia Pacific Simple	B1	\	B3	\	G1900	\	\	B5	B8	\	\	B40	B38/B41
Asia Pacific	B1	\	B3	\	G1900	B7	\	B5	B8	\	\	B40	B38/B41
Overseas Full Band	B1	\	B3	B20	G1900	B7	\	B5	B8	B28A	B28B	B40	B38/B41

Title	Page Name = 65_TRX_MHB	REV: V10	
DOCUMENT NO.:	Design Name = 98860_1_12_201905111250	Size D	
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Friday, May 10, 2019	Sheet 44 of 65	

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LABORATORIO TÉCNICO
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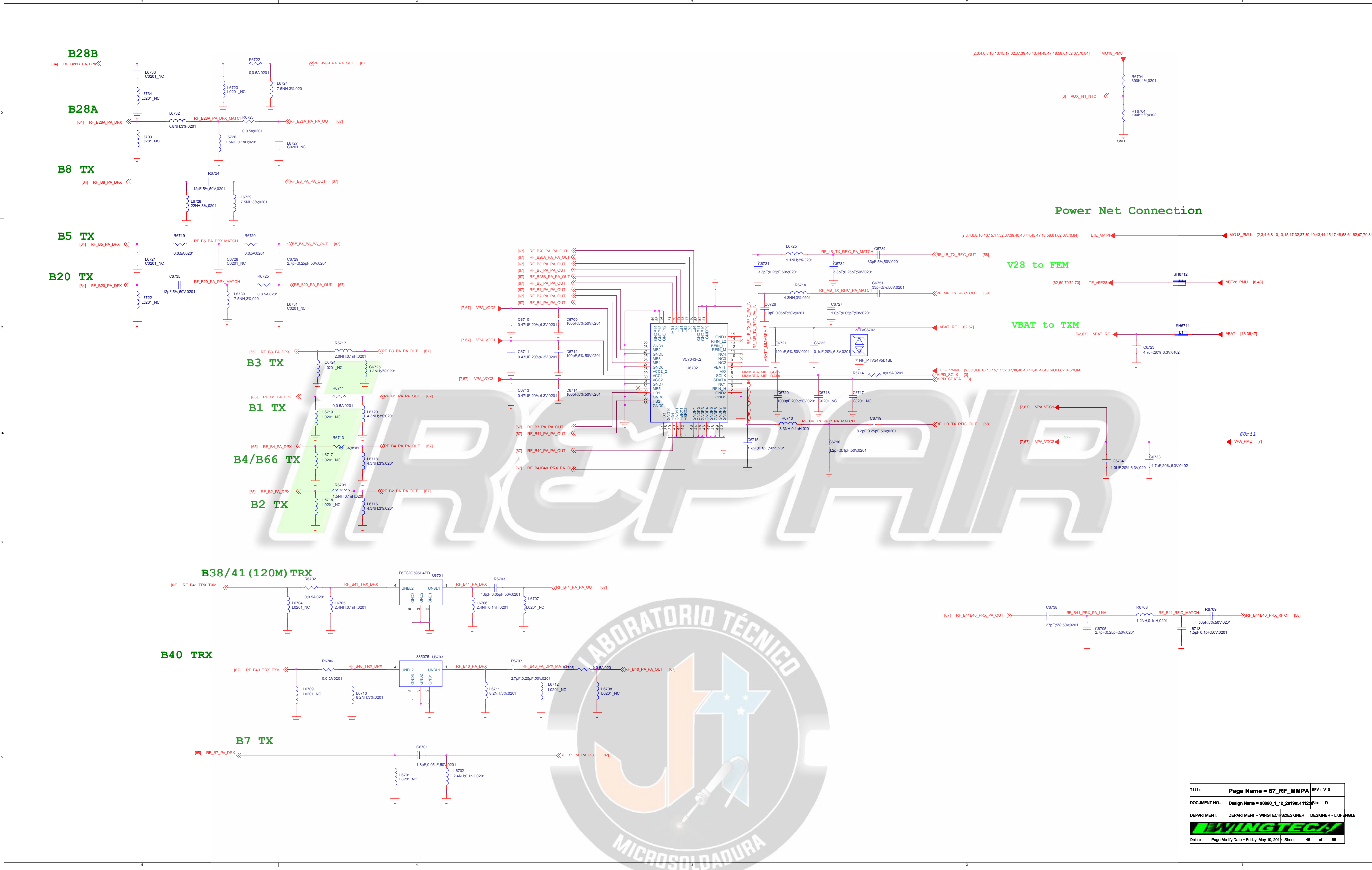
Title <Title>		
Size A1	Document Number <Doc>	Rev 1.0
Date: Friday, May 10, 2019	Sheet 45 of 65	


B28B
B28A
B8 TX
B5 TX
B20 TX
B3 TX
B1 TX
B4/B66 TX
B2 TX
B38/41(120M)TRX
B40 TRX
B7 TX
Power Net Connection
V28 to FEM
VBAT to TXM
VC7643-62
U6702
F6FC2G595H4PD
U6701
885075
U6703
LTE_VMIPI
VIO18_PMU
VPA_VCC1
VPA_VCC2
VPA_PMU
VBAT_RF
VBAT
LTE_VFE28
VFE28_PMU
AUX_IN1_NTC
GND
Title
Page Name = 67_RF_MMPA
REV: V10
DOCUMENT NO.:
Size
D
DEPARTMENT:
DEPARTMENT = WINGTECH-SZ
DESIGNER:
DESIGNER = LIUFENGLEI
WINGTECH
Date:
Page Modify Date = Friday, May 10, 2019
Sheet
46
of
65

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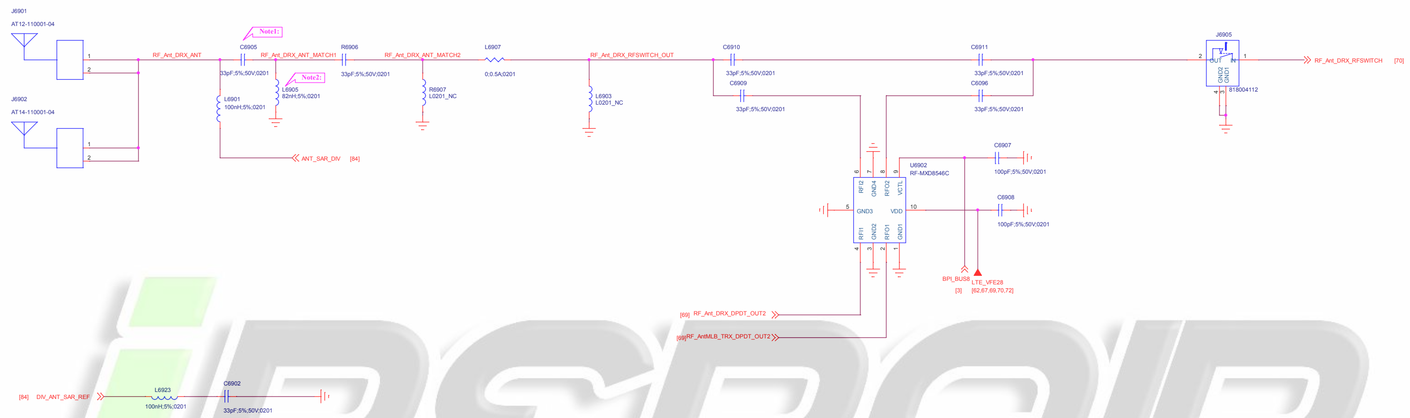
MICROSOLDADURA

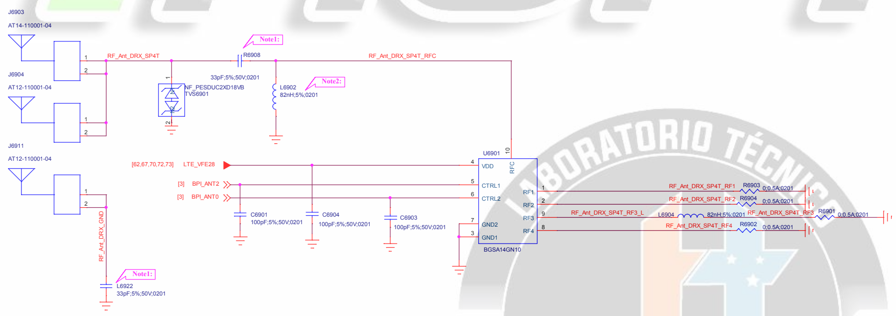
A

Title <Title>		
Size A1	Document Number <Doc>	Rev 1.0
Date: Friday, May 10, 2019	Sheet 47 of 65	

ANT_DIV_FEED



ANT_DIV_TUNER



BGSA14GN10

CTRL1	CTRL2	Operating mode
L	L	RF1 TO RFC
L	H	RF2 TO RFC
H	L	RF3 TO RFC
H	H	RF4 TO RFC

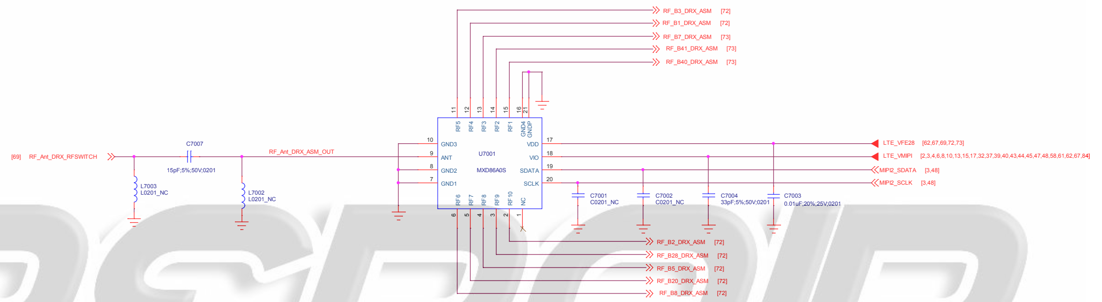
Note1: MUST BE CAP FOR SAR DETECT DESIGN

Note2: MUST BE IND FOR SAR DETECT DESIGN

Note3: 0R RES IS RESERVED FOR SAR SENSOR DESENSE,PLACE CLOSE TO ANTENNA

Note4: share pad in PCB

Title	Page Name = 69_Diversity_ANT	REV	V10
DOCUMENT NO.:	Design Name = 98860_1_12_20190511125	Size	D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Friday, May 10, 2019	Sheet	48 of 65



Title	Page Name = 70_Diversity ASM	REV:	V10
DOCUMENT NO.:	Design Name = 98860_1_12_201905111250	Size	D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Friday, May 10, 2019	Sheet 49 of 65	

WINGTECH

5 4 3 2 1

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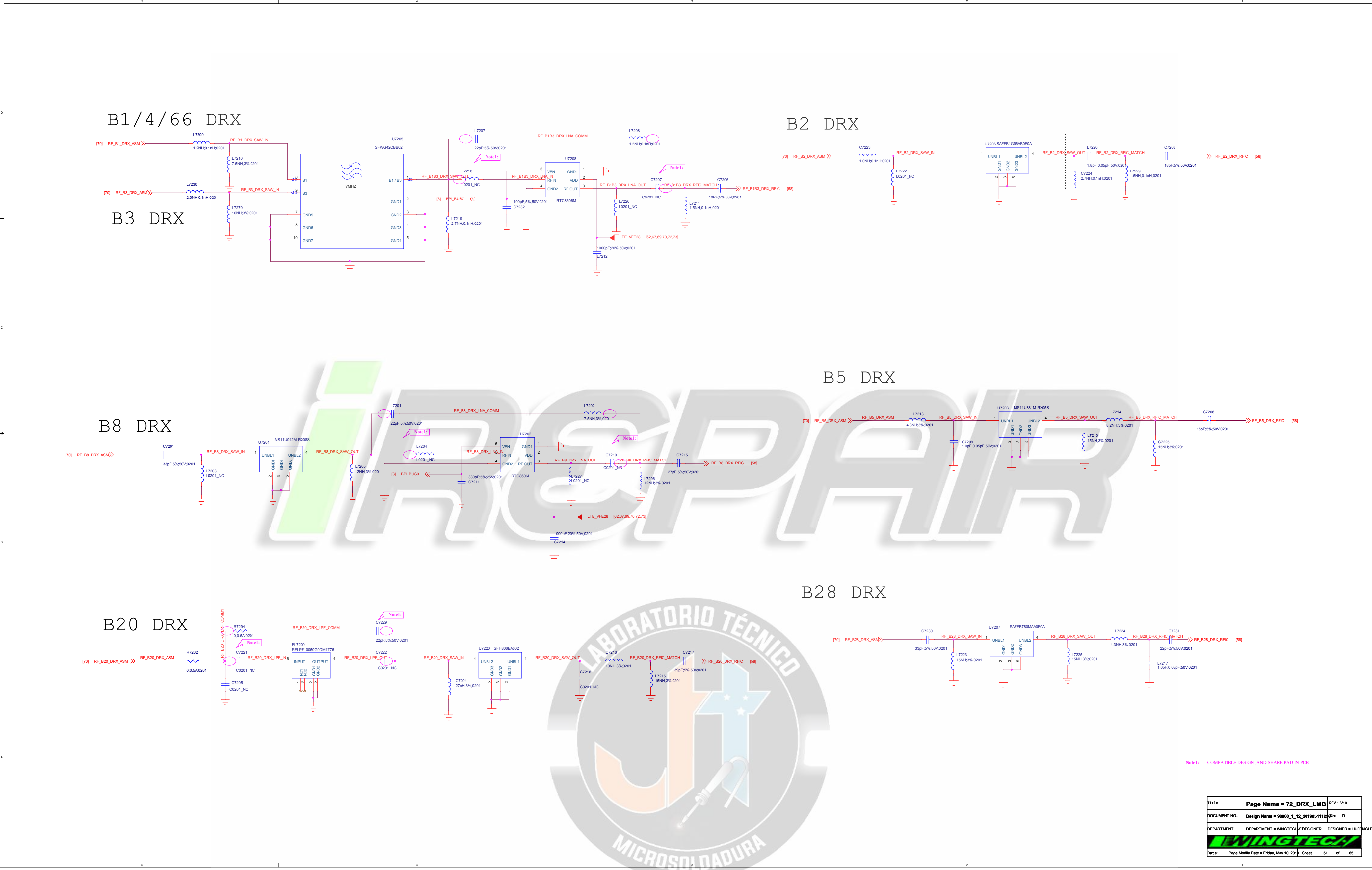
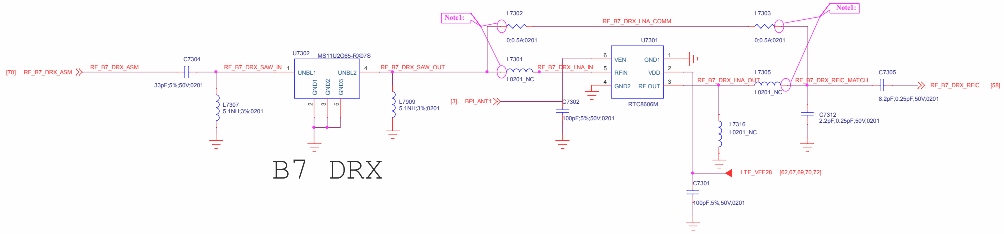
MICROSOLDADURA

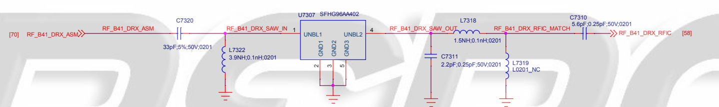
A

Title	Reserved	REV:	V10
DOCUMENT NO.:	Design Name	Size	A1
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Friday, May 10, 2019	Sheet	50 of 65

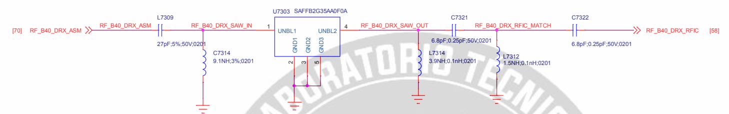

B1/4/66 DRX
B3 DRX
B2 DRX
B5 DRX
B8 DRX
B20 DRX
B28 DRX
Note1: COMPATIBLE DESIGN ,AND SHARE PAD IN PCB
Page Name = 72_DRX_LMB
REV: V10
WINGTECH
Sheet 51 of 65



B7 DRX



B38/41 DRX



B40 DRX

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MICROSOLDADURA

Note1: COMPATIBLE DESIGN ,AND SHARE PAD IN PCB

Title	Page Name = 73_DRX_HB	REV: V10	
DOCUMENT NO.:	Design Name = 98860_1_12_201905111250	Size	D
DEPARTMENT: DEPARTMENT = WINGTECH-SZ		DESIGNER: DESIGNER = LIUFENGLEI	
WINGTECH			
Date: Page Modify Date = Friday, May 10, 2019		Sheet 52 of 65	

5 4 3 2 1

D

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B

A

Title	<Title>	
Size A1	Document Number <Doc>	Rev 1.0
Date: Friday, May 10, 2019	Sheet 53 of 65	

5 4 3 2 1

5 4 3 2 1

D C B A

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DOCUMENT NO.:	Design Name	Size	A1
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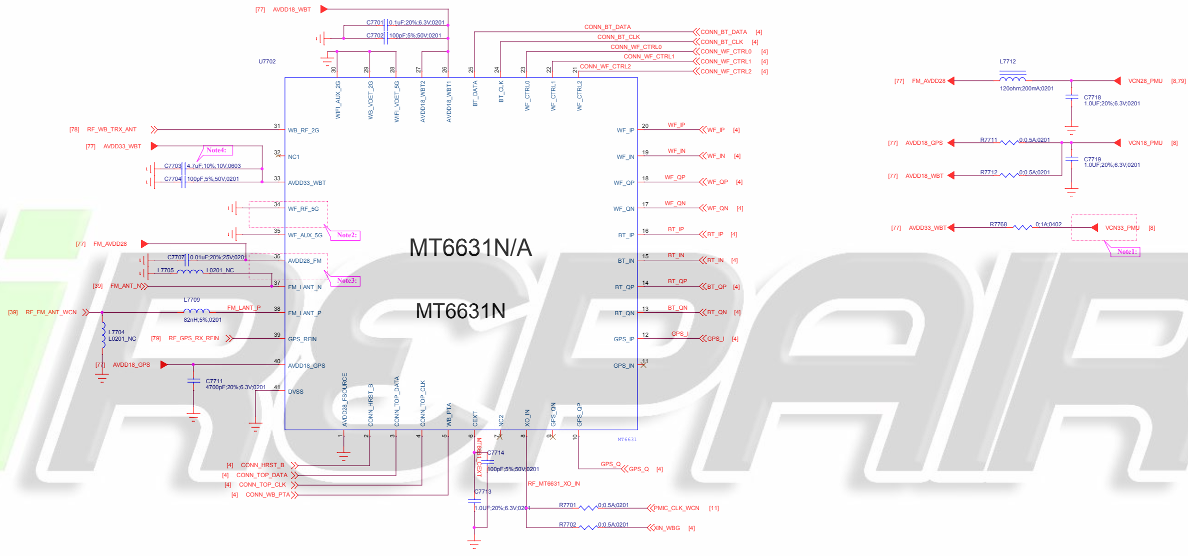

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Note1: If WiFi 5G were no need, VCN33 could be chosen from PMIC output (VCN33_PMU)

Note2: If WiFi 5G not support, connect pin 34(WF_RF_5G) to GND

Note3: Pin 36 (AVDD28_FM) must be connected to VCN28 even if FM not support

Note4: 0603 package use 802167100028

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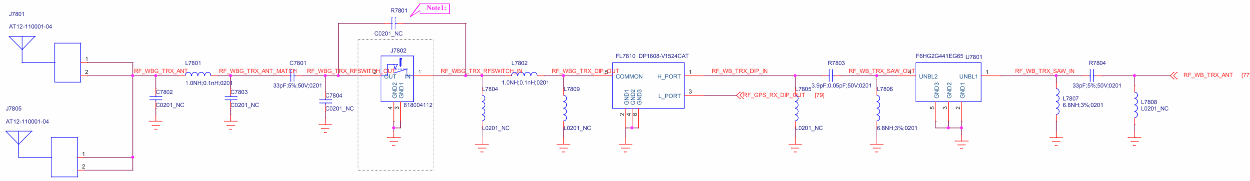
WIFI GPS ANT



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Note1: COMPATIBLE DESIGN FOR WBG RF SWITCH, USE 0R AFTER MP

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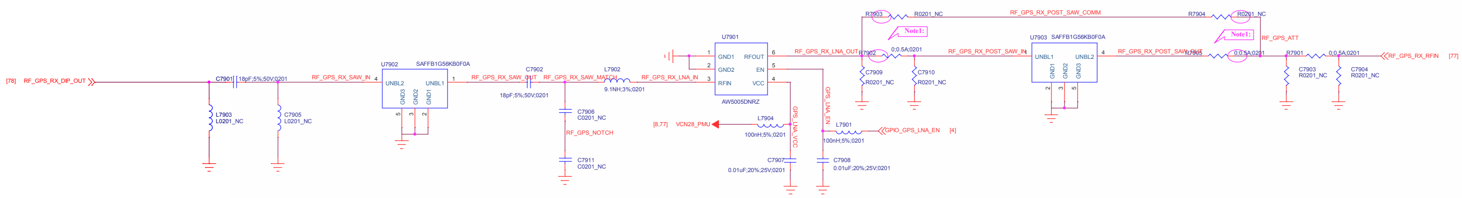

Note1: COMPATIBLE DESIGN FOR GPS POST SAW,SHARE PAD IN PCB

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If SYS CLKis used, the SYS_CLK_REQ signal needs to be connected to the host GPIO pin that configures as SRCLKENAI mode and this pin must be default low

NOTE6:

C8218/C8219 place close to XIN/XOUT Balls (less than 5mm)

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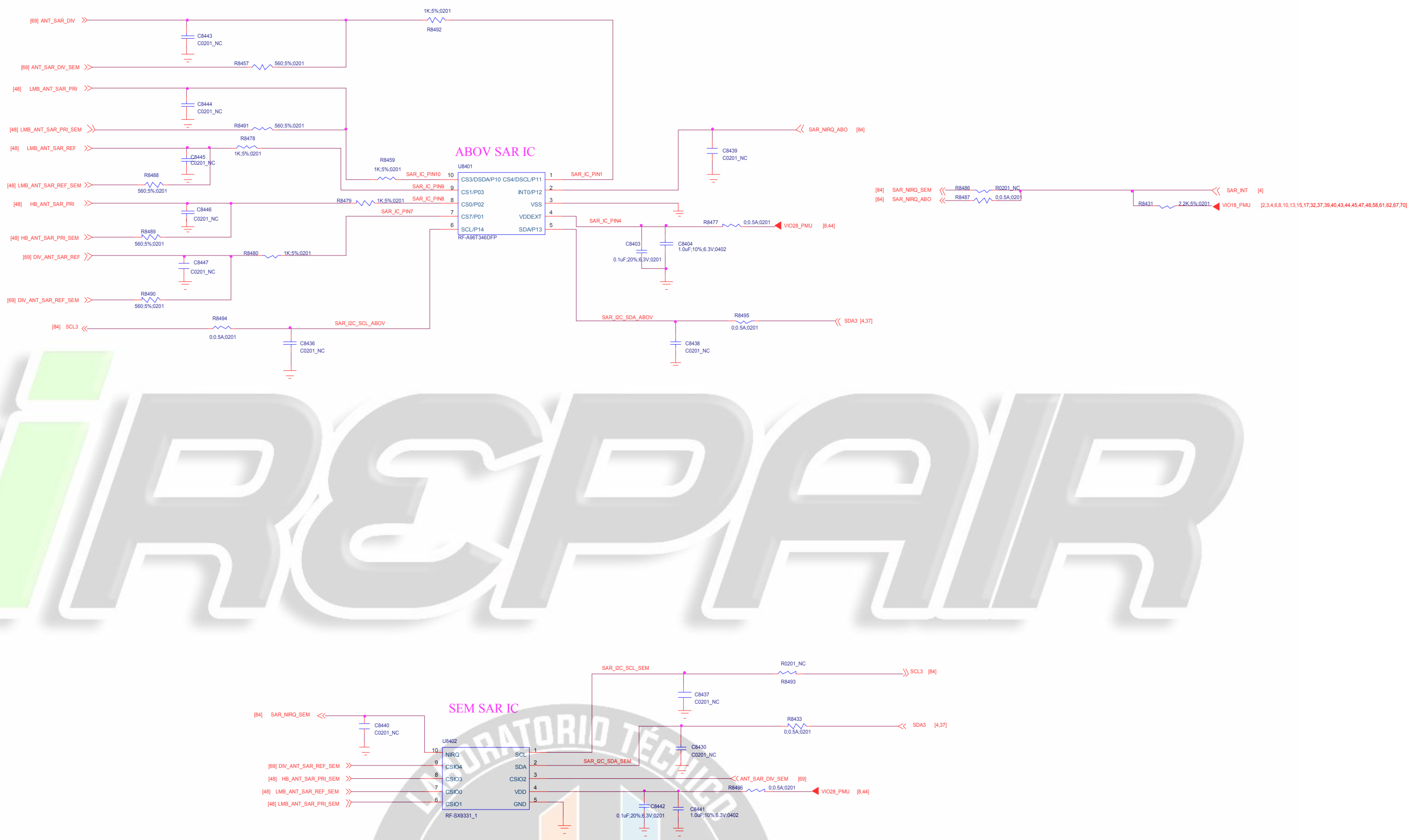

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